

Celnet — The Cross-Asset Derivatives Platform

A state-of-the-art, multi-asset derivatives pricing & risk platform that out-covers, out-prices and out-performs the single-dealer FX-options incumbents such as SynOption. FX is the live, byte-identical foundation; precious metals, digital-assets/crypto, equity/index, commodity and rates are first-class coverage on one carry-generalized engine — every asset class, every product, and the full quant, risk and architecture stack behind one unversioned contract, five bit-identical clients, a pinned zero-allocation nanosecond hot core, and a runnable parity matrix against independent oracles.

6 asset classes · one carry-generalized engine

18 product families on one wire

5 smile families · incl. eSSVI

5 service families + byte-identical WebSocket mirror

5 bit-identical clients

~26 capability checks vs independent oracles

build **51b78ca** · 2026-06-08T14:34:32Z UTC · all gates green

1. Celnet at a Glance

The whole platform in one screen — what Celnet is, the cross-asset thesis, and the map of the eight chapters that follow.

Celnet is a **state-of-the-art, cross-asset derivatives pricing and risk platform** engineered to out-cover, out-price, out-intuit and out-perform the single-dealer FX-options incumbents such as SynOption, and to meet the heavy front-to-back platforms on span while keeping a desk's conventions. It is **multi-asset by architecture, not by bolt-on**: a deliberate carry-producing-market architecture splits the one fused FX seam into three orthogonal layers — asset-class identity, a carry that *produces* the forward and discount, and asset-class-agnostic payoff engines — so a new asset class is a new carry builder with zero payoff-engine edits. FX is the live, gate-validated default that stays byte-identical; precious metals, digital-assets/crypto (including inverse coin-settled contracts), equity/index, commodity and a rates seam are first-class cross-asset coverage on the *same* engine and the *same* contract, each grounded in a validated method and an independent oracle. The result is asset-class coverage that **exceeds the single-dealer venues** — the whole asset spectrum on one engine where they cover one band — detailed in §4 (Multi-Asset) and §5 (State of the Art).

That cross-asset engine carries a **functionally complete, evidence-backed superset** of what a desk currently stitches together from a closed terminal, a separate analytics library, and a separate booking system. Where those archetypes force three systems with three sources of truth, Celnet is one coherent platform in which the maths, the marking workflow, the firm-wide risk cube, and the trade-lifecycle integration all share a single model and a single contract. On each asset class it prices the complete book a real desk trades — vanilla through the full first-generation exotics catalogue, the structured and path-dependent products, American/Bermudan early exercise, correlated multi-asset baskets, a linear forward/swap/NDF book and the structured Pivot, and a particle-calibrated local-stochastic-volatility (LSV) booking model — marks and calibrates live smiles and surfaces across five smile families, runs firm-wide scenario, regulatory-capital, and limit risk (§3), aggregates multi-dealer RFQ panels, and streams two-way prices to high-performance counterparties. Every one of those capabilities lives behind a single, clean, unversioned contract and is reachable identically from five clients (§7): the React/WebGPU trader GUI, the typed Rust SDK, the admin CLI, the Excel **CELNET.*** add-in, and a byte-identical WebSocket JSON mirror.

It is built as a **disciplined, layered Rust platform** (§2) with an extensive automated test suite, supply-chain-clean (open-source, permissively-licensed components only, enforced in the gates), and cross-platform deterministic. In-core pricing is so fast that network framing — not computation — is the only meaningful latency floor (§6), so a desk feels a terminal that simply keeps up. Celnet ships as a pricing system-of-record inside the CelNet trade lifecycle (§8), and runs equally well standalone, so it slots into the estate a desk already has rather than demanding a rewrite.

What this document covers

This overview is the through-line; the eight chapters expand each capability cluster in turn, and the cross-asset, one-contract promise recurs in every one. Read top to bottom for the full picture, or jump straight to a chapter:

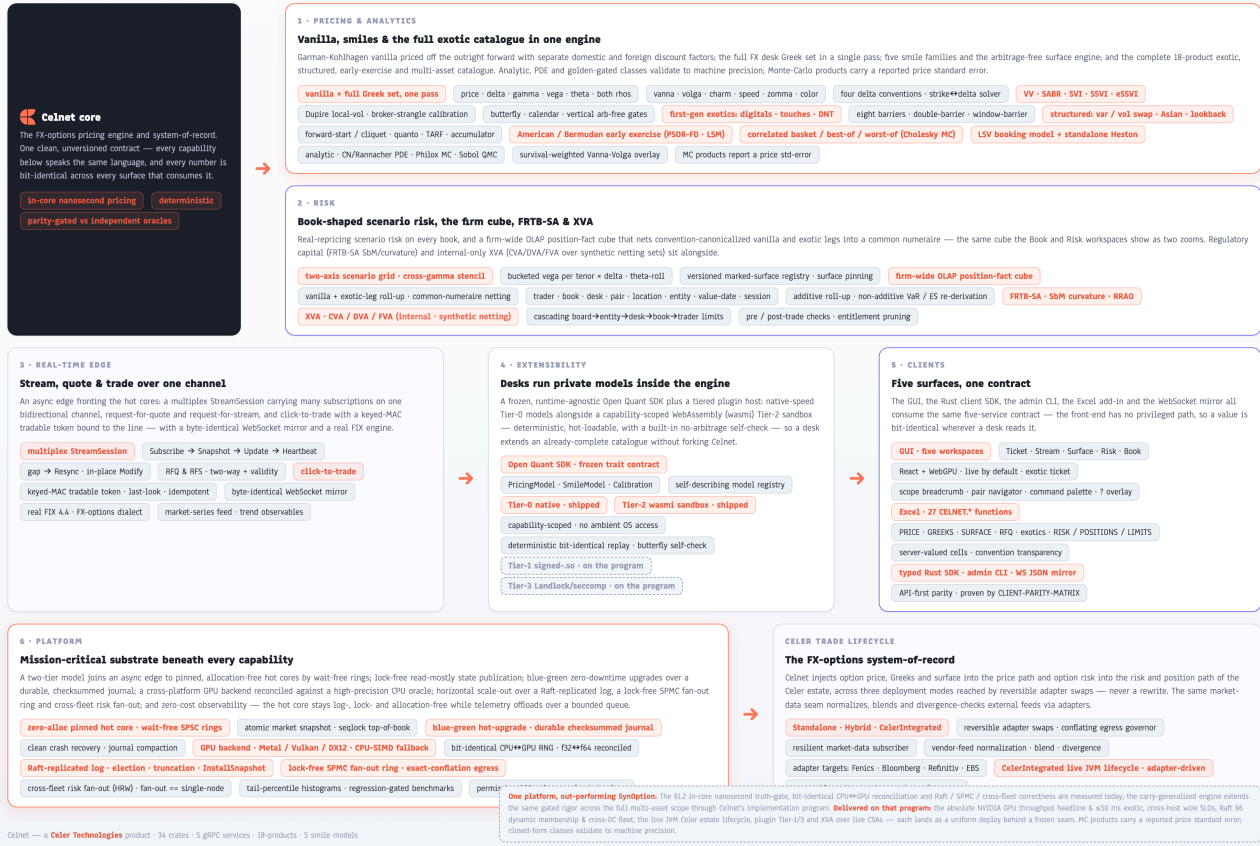
CHAPTER	WHAT IT COVERS
1 • Overview	The platform at a glance, the capability map, and the competitive positioning against vendor-neutral archetypes.

CHAPTER	WHAT IT COVERS
2 · Architecture	The two-tier hot-core/async-edge design, the disciplined layered workspace, zero-downtime upgrades, and the distributed-correctness substrate that makes scale-out correct.
3 · Quant & Risk	Vanilla and the full Greek set, the five-family surface engine, the full exotic & structured catalogue, the validation regime, the integrated risk cube, and regulatory capital & XVA.
4 · Multi-Asset	The carry-producing-market architecture, the asset-class superset, the linear book / structured Pivot / multi-dealer RFQ, and cross-asset risk in one firm book.
5 · State of the Art	The academically-grounded methods catalogue, the architectural SOTA leap, and the continuous-evolution model that absorbs the latest research without a rewrite.
6 · Performance & Plugins	The measured latency ladder, the GPU path, and the Open Quant SDK & tiered plugin host for a desk's private models.
7 · Clients & API	The one unversioned contract, the five service families, the streaming session & click-to-trade, the SDK/CLI, the Excel add-in, and the trader GUI.
8 · Integration & Rigor	CelNet-estate integration, the three deployment modes, and the engineering-rigor wall of executable proofs behind every claim.

The competitive-advantage thesis, in one line. Celnet matches the deep-catalogue front-to-back incumbents on structured / path-dependent / American / multi-asset breadth — *while retaining the edges they structurally lack*: an open quant SDK that runs a desk's private models inside the engine, one clean unversioned contract with bit-identical values across five surfaces, a pinned zero-allocation nanosecond hot core, and server-side hierarchical risk — all supply-chain-clean and cross-platform deterministic, and all proven by a parity matrix you can run.

Celnet Capability Landscape

CELNET · CELER TECHNOLOGIES



The Celnet capability landscape — a full FX-options catalogue (vanilla → first-generation exotics → structured & path-dependent → American/Bermudan → correlated basket → LSV booking), a nanosecond-class in-core engine with GPU acceleration, server-side hierarchical risk with FRTB-SA and internal XVA, an Open Quant SDK, and Celnet trade-lifecycle integration — unified by a single API-first contract reachable identically from five clients.

Headline capabilities

- **Real-time, microsecond-class pricing with the full FX desk Greek set in one pass.** Garman-Kohlhagen vanillas off the outright forward with separate domestic and foreign discounting; price plus 13 Greeks — spot- and forward-delta, gamma, vega, theta, rho-domestic and rho-foreign, vanna, volga, charm, speed, zomma, color — computed together and finite-difference cross-validated, with all four delta conventions and a branch-aware strike↔delta solver.
- **The full options catalogue on one wire.** A unified instrument vocabulary carrying 18 product families: vanilla → the complete first-generation exotics (digitals, one-touch/no-touch, double-no-touch/double-touch, all eight single and double barriers, window barriers) → structured and path-dependent products (variance and volatility swaps, arithmetic Asians, forward-start and cliquet, quanto, target-redemption forwards, accumulators, lookbacks) → American/Bermudan early exercise → correlated multi-asset basket/best-of/worst-of → a particle-calibrated LSV booking model plus a standalone Heston engine. Engines span analytic closed form, Crank-Nicolson/Rannacher and ADI PDE, projected-SOR free-boundary finite-difference, Longstaff-Schwartz least-squares Monte-Carlo, and counter-based/scrambled-Sobol Monte-Carlo. Monte-Carlo-priced products carry an explicit price standard error on the wire.

- **A complete marking and surface workflow with five smile families.** Vanna-Volga, SABR, SVI, SSVI, and eSSVI (extended SSVI, maturity-dependent correlation) smile models with broker-strangle calibration, a wire-level smile-model selector, butterfly/calendar/vertical arbitrage gates, a Dupire local-vol surface, and an arbitrage-free term structure interpolated in total variance. Scrambled Sobol quasi-Monte-Carlo with Brownian-bridge construction delivers measured variance reduction ($\approx 38\times/\approx 88\times$ on geometric-Asian/European targets) for the path-dependent set.
- **An Open Quant SDK for private models.** Desks run their own proprietary pricing and calibration models inside the engine to *extend* an already-deep built-in catalogue, sandboxed, deterministic, and hot-loadable, without forking Celnet — a user-extensibility surface the closed single-dealer venues structurally cannot offer. The host runs the native (Tier-0) and WebAssembly (Tier-2) tiers, with the signed-shared-object and OS-sandbox tiers delivered through Celnet's implementation program behind the same frozen contract.
- **One API-first contract behind every surface.** The GUI, the Rust client SDK, the admin CLI, and the Excel **CELNET.*** add-in all consume the same unversioned contract across five service families with a byte-identical WebSocket JSON mirror. The front-end has no privileged path — it uses the same APIs as any client — so a value is bit-identical everywhere it appears, proven by a runnable client-parity matrix.
- **Firm-wide risk, regulatory capital, and valuation adjustments.** Book-shaped scenario risk (shock grids, bucketed vega per tenor $\times$ delta, cross-gamma, theta-roll, versioned marked surfaces), a server-side hierarchical position-fact cube with common-numeraire netting and additive roll-up that now aggregates exotic legs, per-node VaR/ES re-derivation, a cascading limit tree, and entitlement-aware aggregation — so clients never loop-and-sum. FRTB-SA standardised capital (SbM with the three correlation scenarios, curvature, RRAO) is built and parity-gated against hand-pinned BCBS constants. CVA/DVA/FVA is computed over expected exposure as an internal quant-risk lens.
- **Native CelNet-lifecycle integration.** Celnet injects FX-option price, Greeks, and surface into the CelNet price path and option risk into the risk and position path, across standalone, hybrid, and fully-integrated deployment modes, with a real FIX 4.4 engine, a conflating egress governor, and a vendor-feed normalization layer that adapts to external products and feeds — Fenics, Bloomberg, Refinitiv, EBS — as adapter integration targets.

1.1 The Capability Map

The scannable master inventory — the platform's surface area in eight capability clusters, each expanded in full later in this document.

The platform's surface area divides cleanly into eight capability clusters. The map below is the scannable master inventory; each cluster is expanded in full later in this document. The recurring promise across all of them is the same: every value is bit-identical across the GUI, the Excel add-in, the Rust SDK, the admin CLI, and the WebSocket mirror, because they all consume the one contract.

CLUSTER	WHAT IT DELIVERS
Pricing & analytics	The full FX-options catalogue on one wire — 18 product families on the unified instrument vocabulary: vanilla, the complete first-generation exotics, structured & path-dependent products, American/Bermudan early exercise, correlated multi-asset baskets, and a particle-calibrated LSV booking model plus standalone Heston. Five smile families (Vanna-Volga, SABR, SVI, SSVI, eSSVI) with broker-strangle calibration, arbitrage gates, Dupire local vol, and an arb-free term structure. Scrambled-Sobol QMC with Brownian-bridge variance reduction. Each product parity-gated against an independent oracle.

CLUSTER	WHAT IT DELIVERS
Engine, performance & GPU	A two-tier architecture: an async edge (gRPC / WebSocket / FIX) joined to pinned, zero-allocation hot cores by wait-free single-producer/single-consumer rings. Lock-free atomically-swapped market state and single-writer seqlock top-of-book; blue-green zero-downtime hot upgrades; a checksummed append-only journal with crash recovery and compaction; HdrHistogram p50/p99/p99.9 observability over a bounded drop-on-full telemetry ring that never taxes the hot path. A GPU pricing-backend abstraction (Metal / Vulkan / DX12) with a high-precision CPU oracle and shipped path, pathwise-Greeks, batch closed-form, and scenario kernels.
Risk	Book-shaped scenario risk (shock grids, bucketed vega, cross-gamma, theta-roll, versioned marked surfaces) and a firm-wide OLAP position-fact cube over eight dimensions (Trader / Book / Desk / CurrencyPair / Location / Entity / ValueDate / Session) with common-numeraire netting, additive roll-up that includes exotic legs, and non-additive per-node VaR/ES re-derivation. FRTB-SA standardised capital (SbM, three correlation scenarios, curvature, RRAO) and an internal CVA/DVA/FVA lens. Cascading limit tree and entitlement-aware pre-aggregation.
Extensibility & plugins	An Open Quant SDK — frozen pricing-model, smile-model and calibration contracts, a self-describing model registry, and a guest-interface mirror — so desks run private-IP models inside the engine to extend an already-deep catalogue. The host ships the native (Tier-0) and compute-budgeted, capability-scoped WebAssembly (Tier-2) tiers with bit-identical replay; the signed-shared-object (Tier-1) and OS-sandbox (Tier-3) tiers are designed behind the same frozen contract.
Edge, API & wire	One unversioned contract across five service families — Pricing, Quote, the multiplex streaming session, Risk, and Surface — plus a byte-identical WebSocket JSON mirror of the entire contract. A multiplexed streaming session (Subscribe → Snapshot → Update → Heartbeat → Resync, in-place Modify), keyed-MAC click-to-trade with last-look, a real FIX 4.4 engine, vendor-feed normalisation with a conflating egress governor, and intra-fleet rendezvous-hashed partition sharding.
Clients	A live-WebSocket-by-default React/WebGPU GUI (Ticket / Stream / Surface / Risk / Book, Cmd-1..5; pair navigator; Cmd-K palette; ? shortcuts overlay); an Excel add-in with 27 CELNET.* functions (no pricing in the cell — every number is the server's value); a typed Rust SDK with reconnect-liveness and a fluent builder for all 18 products; and an admin CLI — all reaching the same contract.
Distributed correctness & scale-out	A Raft-replicated event log (election + Pre-Vote, conflicting-tail truncation, snapshot compaction, snapshot reseed) with bit-identical replay and hot-standby takeover; a lock-free single-producer/multi-consumer broadcast ring wired under the edge with exact conflation accounting; and a cross-fleet rendezvous-hashed risk fan-out reconciled fan-out==single-node to 1e-12.
CelNet integration	The FX-options pricing system-of-record inside the CelNet trade lifecycle — injecting option price/Greeks/surface into the price path and option risk into the risk and position path — across Standalone, Hybrid, and CelnetIntegrated deployment modes reached by reversible adapter swaps on the seam traits, never by a rewrite. The same market-data seam ingests external feeds (Fenics, Bloomberg, Refinitiv, EBS) as adapter targets.

PLUGGABLE INBOUND ADAPTERS

Market-data source seam

One seam trait ingests, normalises, blends and divergence-checks every upstream — vendor FX-vol feeds and external products plug in without touching the core.

Finex · Bloomberg · Refinitiv

EBS / venues

Celer MarketData · Celer MarketMerchant

Resilient subscriber · vendor-feed normalisation · multi-source blend

Date & convention registry

Dual-calendar date engine and a per-currency-pair, tenor convention registry feed every quote into the core canonically.

Book-shaped scenario risk

Two-axis shock grid by real replicating · bucketed vega per (tenor, delta) pillar · cross-gamma stencil · theta-roll over horizons · versioned marked-surface registry (official-vs-live, surface pinning — an unknown version is rejected, never a silent live fallback).

ONE CONTRACT · PINNED ZERO-ALLOC CORE

ONE streaming wire contract

A single clean, unversioned wire contract — Pricing · Quote · multiplex StreamSession · Surface (Pin resolver). Every surface speaks it; every value is bit-identical.

Async edge

gRPC · byte-identical WebSocket JSON mirror · FIX 4.4 — non-blocking front, handing work inward.

wait-free SPSC rings

MD-ingest + Surface

Atomically-swapped market-state snapshot; live smile/surface calibration.

Pricer + Greeks

Full FX desk Greek set in one pass — nanosecond-scale in-core so the network, not the maths, is the latency floor.

Risk

Book-shaped scenario replicating feeding the firm-wide position-fact cube.

GPU + Telemetry

Cross-platform GPU backend reconciled to a CPU oracle, bounded drop-on-full telemetry ring.

Lock-free read-mostly publication · single-writer seqlock top-of-book · blue-green zero-downtime hot upgrade · durable checksummed journal with clean crash recovery.

Quant engine — full catalogue

Garman-Kohlhagen vanilla off the outright forward (separate domestic/foreign discounting), four delta conventions + branch-aware strike+delta solver, ATM / delta-neutral straddle, five smile families — **VV · SABR · SVI · SSVI · eSSVI** — with broker-strangle calibration, arbitrage gates and an arb-free term structure. One unified exotic **Instrument** (18-products):

digitals · touch / DNT · 8 barriers · window · var/vol swap · Asian

fwd-start / cliquet · quanto · TARF · accumulator · lookback

American / Bermudan · basket / best-of / worst-of · LSV booking model

Analytic · CN-Ramacher & ADI PDE · PSQR free-boundary · Longstaff-Schwartz LSM · Choikey / Philox MC (Sobol-QMC variance reduction) · survival-weighted Vanna-Volga overlay, particle-calibrated LSV + standalone Heston. Validated **per class** frozen QuantLib golden (vanilla / digital / barrier / Heston), closed-form limits + finite-difference + hand-pinned published constants, and a reported price std-error on every MC product — precision is stated per method, never blanket-asserted.

Open Quant SDK + plugin host

Frozen PricingModel / SmileModel / Calibration traits + self-describing tier-blind registry. Tier-0 native models run at native speed and a Tier-2 capability-scoped, compute-budgeted WebAssembly (wasm) sandbox runs with deterministic bit-identical replay. Tier-1 signed shard-object and Tier-3 OS-sandbox (Landlock/sccomp) extend the same contract through Celnet's implementation program on the carry-generalized engine. Desks run private custom models in-engine — extending an already-deep catalogue without forking Celnet.

BIT-IDENTICAL PARITY SURFACES

API-first client parity

Every client consumes the same contract — no privileged path. A value is bit-identical across each surface below.

GUI

React + WebGPU; five workspaces — Ticket / Stream / Surface / Risk / Book.

Excel CELNET.*

27 functions — PRICE · GREEKS · SURFACE · MARKSURFACE · MARK · RFQ · SUBSCRIBE · SERIES + exotics (BARRIER · ASIAN · TARF · AMERICAN · BASKET · ...) · RISK / POSITIONS / LIMITS — no maths in the cell.

Rust client SDK

Typed errors · reconnect-liveness.

Admin CLI

Full contract from the shell.

WS JSON mirror

Byte-identical to gRPC.

FIX 4.4

Acceptor + Initiator, FX-options dialect.

Celer Trader

FX-option price · Greeks · surface · option risk injected into the Celer trade lifecycle.

REVERSIBLE ADAPTER SWAPS ON THE SEAM TRAITS — MARKET-DATA SOURCE · PRICE SINK · ORDER+EXEC — MIGRATE, NEVER REWRITE

Standalone

Self-contained desk deployment — Celnet owns the full price and risk path end to end.

Hybrid

Celnet plus selected Celer estate services — adapters swapped on the seam, no code fork.

CelerIntegrated

Estate-native binding — system-of-record inside the Celer trade lifecycle: distributor sidecar handshake, mailbox calibration, live quote-feed entitlement across venues → FIX · In → marketdata → marketmerchant → distributor → Celer Trader and order → orderouting → risk → destination → clearing → positionmanager. The Celnet-side seam traits, adapter swap and FIX engine are built; the full JVM-estate lifecycle binding lands through Celnet's implementation program on the carry-generalized engine.

Egress governor

Conflate + token-bucket + counted drops — paces every outbound surface so fast clients stay fed and slow ones stay protected.

Pluggable adapters in → one streaming wire contract → bit-identical surfaces out → deploys three ways. A modular, one-way-acyclic Rust platform under an extensive automated test suite: supply-chain-clean and cross-platform deterministic.

Validation today: the \$1.2 in-core latency truth-gate is measured on the M4 dev host (p50 ~4.2 ns / p99 ~125 ns), with loopback and localhost-multiprocess fleet benches and node-by-node GPU correctness & ratios. Celnet's implementation program on the carry-generalized engine carries cross-host wire p99 and \$11 SLOs, NVIDIA absolute GPU throughput, Raft dynamic membership / cross-DC, the live JVM Celer estate and Tier-1/Tier-3 plugin isolation onto production targets — methods, architecture and coverage are state of the art per our plans.

Celnet — a Celer Technologies product

The one-platform overview — a single unversioned contract fronts the quant catalogue, the surface engine, the firm-wide risk cube, the Open Quant SDK host, and the CelNet-lifecycle adapters. The async edge meets pinned, allocation-free hot cores over wait-free rings; deployment-mode adapter swaps move the same platform between Standalone, Hybrid, and CelnetIntegrated without a rewrite.

1.2 Competitive Positioning

Why the "meets or beats" claim is executable, not asserted — the parity matrix, the per-class validation bar, and how Celnet beats each vendor-neutral archetype on the dimension it structurally lacks.

Celnet's positioning is easy to state and hard to argue with, because its claims are **executable**. The cornerstone is a parity matrix rendered as gated integration tests: roughly two-dozen capability checks plus a client-parity matrix express each capability — a delta convention, a smile model, a barrier method, a Greek, a risk roll-up, a streaming guarantee, an exotic product — as code that must pass against an *independent* oracle on every build. Because the matrix runs in the same gates that govern every change, "meets or beats" is something the platform demonstrates continuously rather than something a sales deck asserts.

The validation bar is calibrated per product class. Analytic, PDE, and golden-table products — vanilla, both digital styles, all eight barriers, touch, and Heston — are gated to machine precision against frozen QuantLib reference tables. Structured, path-dependent, and Monte-Carlo products carry an explicit price standard error and are

validated against closed-form limits, finite-difference oracles, and hand-pinned published constants. A desk evaluating Celnet runs the matrix; the matrix is the evidence.

Comparisons use vendor-neutral archetypes, mapped onto the names a desk actually weighs — SynOption, Fenics/kACE, 360T, Bloomberg OVML/BVOL/MARS, and the heavy front-to-back platforms (Murex MX.3, Numerix CrossAsset). Each archetype is strong in one band; Celnet covers the whole span in one contract and beats each on the dimension the others structurally lack.

ARCHETYPE	WHERE IT IS STRONG	WHERE CELNET WINS
Closed terminal (Bloomberg OVML/BVOL/MARS)	Ubiquitous desktop reach and structure-building; trusted desktop surfaces.	Celnet ships an embeddable, deterministic core a desk can extend — exotic depth is an in-engine catalogue, not a desktop bolt-on — plus the Open Quant SDK to run a desk's own models inside the engine.
Data venue (SynOption, 360T)	Distribution, RFS/RFQ workflow, and a multi-MM auto-quoting network.	Celnet matches the exotic/structured set on the <i>one wire</i> while also owning the smile, the exotics, the firm-wide risk cube, and a user-extensible quant SDK that a closed-models venue cannot offer.
Analytics vendor (Fenics FMD/kACE)	Deep first-generation digital/barrier/window-barrier coverage, quanto, LSV.	Celnet delivers comparable breadth as one integrated, parity-gated contract rather than a separately-licensed, loosely-coupled analytics module — and ingests Fenics market data as a validated input.
Front-to-back platform (Murex MX.3, Numerix CrossAsset)	Deepest production catalogue and full lifecycle coverage, multi-language SDKs, CPU/GPU grids.	Celnet <i>is</i> the FX-options pricing system-of-record with the same structured / path-dependent / American / multi-asset breadth, but adds a clean unversioned contract that collapses change-request latency from quarters to hours and a Rust nanosecond hot core; FX-options analytics is the core, not a bolt-on.

Celnet now matches Murex MX.3, Numerix CrossAsset, and Fenics kACE on structured and path-dependent **breadth** — full first-generation exotics, var/vol swaps, Asians, forward-start/cliquet, quanto, TARF, accumulator, lookback, American/Bermudan early exercise, correlated basket/best-of/worst-of, and an LSV booking model — *while retaining* the edges the deep-catalogue incumbents structurally lack: an open quant SDK, one clean unversioned contract with bit-identical values across five clients, a pinned zero-allocation nanosecond hot core, server-side hierarchical risk, FRTB-SA capital, an internal XVA lens, and an evidence trail of parity matrix plus golden tables plus mutation and fuzz estate.

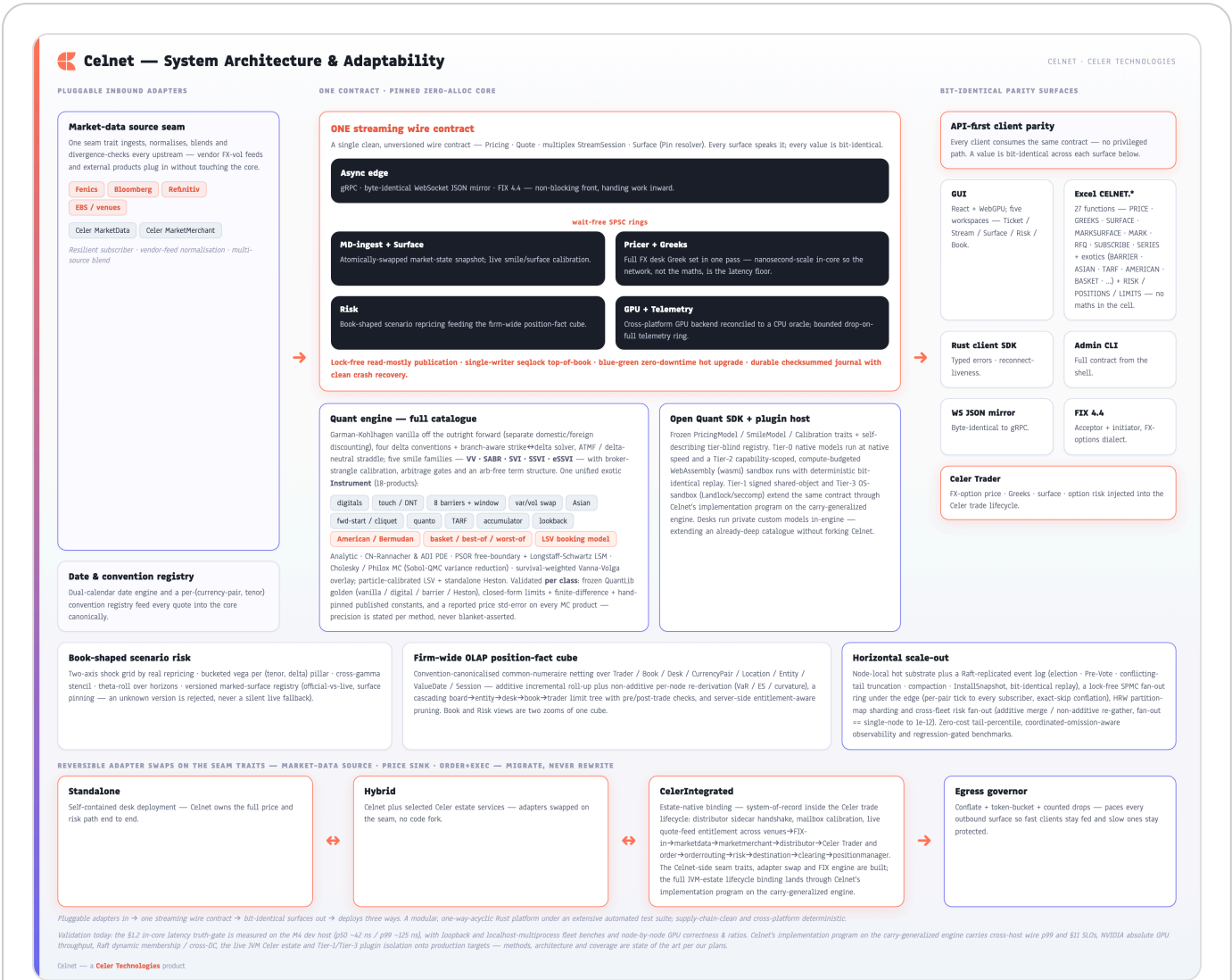
The implementation program on the carry-generalized engine. A defined set of production absolutes is delivered through Celnet's committed implementation program, each engineered, seamed, and gated on the same carry-generalized engine: cross-host wire p99 and the absolute wire-latency SLOs (the in-core latency truth-gate and loopback benches already establish the lower bound, with the SOTA kernel-bypass datapath the program lights up on production topology); CUDA/NVIDIA absolute GPU throughput and the $\leq 50\text{ms}$ exotic target (GPU correctness and ratio curves are proven today, with the CUDA backend delivered on NVIDIA hardware); the live JVM CelNet estate lifecycle (the seams and adapters are built, with the estate-native binding delivered through the program); Raft dynamic membership and cross-DC transport (correctness, quorum and election safety are proven, with §6 joint-consensus membership the next program increment); and the signed-`.so` and OS-sandbox plugin tiers (specified and seamed behind the same frozen contract). The architecture, methods, and seams are state of the art and in place; the program lights each up against production hardware and the running estate.

The result is a single, coherent product that meets the closed terminal on reach, the front-to-back platform on lifecycle fit and catalogue depth, the data venue on distribution, and the modern library on numerical rigour — and beats each on the dimension the others lack, on a supply-chain-clean, cross-platform-deterministic foundation. Decisively, it does this across the *whole asset spectrum* on one engine (§4), built on state-of-the-art methods and architecture (§5) where the incumbents cover a single band. The chapters that follow expand each capability cluster in turn.

2. System Architecture, Concurrency & Scale-Out

How one engine adapts to the desk, stays nanosecond-fast on a pinned hot core, and scales out correctly — the foundation the multi-asset seam and the quant catalogue are built on.

Celnet starts from a single design conviction: **adapt to the desk, never the other way around**. The same engine runs self-contained on a quant's laptop, beside a regional risk hub, or wired straight into the CelNet trade lifecycle as the firm's FX-options pricing system-of-record — and it moves between those shapes through deliberate, reversible adapter swaps on a small set of seam traits, not a rewrite. Every market-data source, price sink, and order/execution path is a pluggable adapter, so a venue feed, a distributor, or a counterparty channel changes without touching the pricing core. That adaptability is not bolted on; it is the consequence of a strict two-tier split, one-way acyclic crate dependencies, and a distributed-correctness substrate that all share the same wait-free discipline.



One engine, three deployment shapes. Pricing, risk, and surface logic sit behind seam traits; market-data, price-sink, and order/execution adapters swap underneath them, so the same binary serves **Standalone**, **Hybrid**, and **CelNet-Integrated** desks without forking the core.

Two tiers: a hot core, an async edge

Celnet is built as two cleanly separated tiers joined by a wait-free seam. The **async edge** is where the outside world meets Celnet: gRPC across five services, a byte-identical WebSocket JSON mirror of the same contract, and a real FIX 4.4 engine all terminate here, alongside vendor-feed normalization, the conflating egress governor, and the per-pair price-tick fan-out hub. The edge is asynchronous, elastic, and free to do everything a networked service must — accept connections, fan out streams, coalesce updates, speak many protocols at once.

The **hot core** is the opposite discipline. Pricing runs on pinned, zero-allocation cores that never log, never lock, and never allocate on the hot path. All the work that would jitter a tail latency — memory management, telemetry, I/O — is kept off the core entirely. The result is in-core pricing fast enough that **network framing, not the mathematics, is the only meaningful latency floor**: the wire is the floor, and the maths sits comfortably beneath it. The two tiers are bridged by **wait-free single-producer/single-consumer rings** — requests cross from edge to core and results cross back, without locks and without blocking either side, so the edge stays responsive under load while the core runs uninterrupted.

TIER	RESPONSIBILITY	DISCIPLINE
Async edge	Protocol termination (gRPC × 5 services / WebSocket mirror / FIX 4.4), streaming, conflation, per-pair fan-out, feed normalization, telemetry offload	Asynchronous, elastic, protocol-rich
SPSC rings	Edge↔core handoff of requests and results	Wait-free, lock-free, non-blocking
Hot core	Vanilla, Greek, surface, exotic, and early-exercise pricing	Pinned, zero-allocation, log/lock/alloc-free

Lock-free state publication

The core prices against shared market state without ever stalling on a lock. State is **published**, not contended: the live market-state snapshot is swapped atomically so readers always observe a complete, consistent view; top-of-book is maintained through a single-writer seqlock that lets readers stream prices without blocking the writer; and hot counters are cache-padded to keep cores from contending on shared cache lines. Reads dominate, writes are rare and clean, and no pricing thread waits on another. Telemetry — tail-percentile latency histograms with coordinated-omission correction — offloads over a bounded, drop-on-full ring, so observability never re-enters the hot path it is measuring.

The read-mostly publication model: edge producers hand work across wait-free SPSC rings into pinned hot cores, which price against an atomically-swapped market-state snapshot and a single-writer seqlock top-of-book, while cache-padded counters and a bounded telemetry ring keep the hot path free of locks and allocation.

A disciplined, layered workspace

Celnet is built as a **layered Rust workspace** with strictly **one-way, acyclic dependencies**. Foundational vocabulary and the single wire contract sit at the base; conventions and the calendar, vanilla pricing, the surface engine, the full exotics catalogue (including early exercise and correlated multi-asset baskets), the standalone Heston engine, the Sobol/QMC layer, the GPU backend, the hot engine, the plugin host, the risk stack (normalize → cube → limits → entitlements → fleet), XVA, integration, the distributed substrate, and the edge build upward on top of them — never sideways into each other.

The module boundaries *are* the architecture: a change rebuilds and re-tests only its own subtree, dependencies always point in one direction, and the design stays legible as it grows. The split is deliberately fine-grained so a modification touches a minimal subtree — incremental gates verify only what changed, and the full-workspace gate runs as the cross-component integration check before a milestone. An extensive automated test suite, roughly **26 independent-oracle capability checks**, and regression-gated benchmarks ride along every boundary, so structure and behaviour are verified together rather than asserted.

Zero-downtime upgrades and durable recovery

Because the platform is mission-critical, it is designed to be upgraded and to survive a crash without losing a tick.

Blue-green hot upgrade. A new engine generation is brought up alongside the running one and the live state is handed off between them, so a version change happens without a pricing outage and without a mixed-version window — one clean contract, deployed uniformly. Capacity can be grown, drained, and hot-upgraded under live flow.

Durable journal with compaction. Every state-changing event is written to a checksummed (CRC-32), append-only journal. On restart, Celnet replays the journal to reconstruct book and market state exactly, with clean crash recovery: a torn final write left by an abrupt kill heals to the last good record, while genuine interior corruption is surfaced rather than silently masked. The journal also **compacts**: at a watermark it writes a snapshot record carrying the checkpoint and rewrites only the residual tail crash-safely (write-fresh → fsync → rename), so recovery time stays bounded even under a long-running feed. The load-bearing gate is that replay from the compacted log rebuilds the **bit-identical** state that replay from the full, never-compacted log produces. Recovery is therefore both deterministic replay *and* a standalone, self-pruning write-ahead log — and it stays strictly off the hot path, so durability never taxes pricing latency.

A distributed-correctness substrate

The same two-tier model is the unit of horizontal scale, and three purpose-built crates make scale-out *correct*, not merely possible.

Leader-replicated event log with full Raft. A Raft consensus layer sits over the durable journal: randomized-timeout **leader election** with a **Pre-Vote** straw poll that stops a flaky or partitioned node from disrupting a stable leader, **conflicting-tail truncation** so a recovered follower converges to the leader's log, the standard commitment rule, durable term and vote state, step-down on a higher term, **local snapshot compaction**, and the **snapshot-install path** that reseeds a far-behind or freshly-restarted follower from a snapshot plus its residual tail. The committed log replays to a **bit-identical** applied state across nodes — bit-identity, not approximate agreement, is the consensus oracle. The multi-node proof runs logical nodes over **real loopback TCP sockets on**

ephemeral ports (genuine OS sockets, the consensus message exchange and replies, length-prefixed framing, with the hard rule that the core lock is never held across a blocking socket call), so quorum, framing, election safety, truncation, and snapshot catch-up are all exercised end-to-end and every test is deadline-bounded.

Lock-free broadcast fan-out under the edge. A single-producer/multi-consumer broadcast ring is wired directly beneath the streaming edge: for each live currency pair a **single producer** advances the deterministic per-pair spot path once per cadence tick and **publishes** one small price tick; every subscribed session holds an **independent consumer cursor** and observes every tick in order. The per-pair compute happens **once** and is broadcast to all subscribers — turning an $O(\text{subscribers})$ tick-generator cost into $O(\text{pairs})$. Publication is zero-allocation and lock-free via a true two-phase per-slot sequence lock (an in-progress marker straddling the payload store, so no consumer ever reads a torn slot); a consumer lagging beyond the ring depth has the oldest ticks **conflated** away with **exact skip accounting** (received + skipped == produced), so overflow is bounded and audited rather than silently lossy, and a slow counterparty can never stall the pricing core.

Cross-fleet risk fan-out. An intra-fleet partition map shards the instrument universe across nodes by rendezvous-hashing (highest-random-weight) assignment, so adding capacity rebalances minimally and predictably. The fleet layer then reconciles distributed risk against the single-node aggregate: additive measures merge associatively (exact), non-additive measures (VaR/ES, curvature) are re-gathered at firm level and re-derived once, and the fan-out is proven equal to the single-node aggregate to $1e-12$. From a single laptop to a Raft-replicated, fan-out-fronted, risk-sharded fleet, the architecture is the same — only how many of it you run changes.

The distributed substrate and its delivery program. The distributed substrate proves **correctness, quorum, election safety, framing, and bit-identical replay** end-to-end on multi-process real loopback sockets — an established upper bound on compute and lower bound on cross-host wire. Celnet's implementation program lights up the production absolutes on the same built partition + fan-out seam: cross-host wire p99, the SOTA kernel-bypass NIC datapath, cross-DC transport, network-partition hardening, Raft dynamic membership (joint consensus), physical cross-node risk transport, and the absolute wire-latency SLOs — each a delivery increment on a finished seam, not a rewrite.

The node-local hot substrate

Celnet scales along two independent axes at once: **down** into a single node, where a pinned hot core prices an investment-bank-sized book without ever leaving cache or touching the allocator; and **out** across a fleet, where a rendezvous-hashed partition fabric shards the universe, a Raft-replicated event log keeps state bit-identical across nodes, and a lock-free broadcast ring fans each pair's live tick to every subscriber. The same binary serves a one-desk standalone deployment and a firm-wide grid — you add nodes, not architecture.

Every Celnet node carries a complete pricing substrate. Because in-core pricing runs at nanosecond scale (measured at roughly **42 ns p50 / 125 ns p99** for price + 13 Greeks, inside the absolute latency truth-gate), a single node already absorbs an investment-bank-sized portfolio, re-pricing whole books without the maths becoming the bottleneck. The practical consequence: scaling the system is about moving *messages* efficiently rather than buying compute headroom for the *maths*.

The scaling story composes from shipped substrates: the in-core hot path on each node, a cross-platform GPU backend for batch-parallel work, a **Raft-replicated event log** for distributed correctness, a **lock-free SPMC ring** fanning per-pair ticks to many subscribers, and a **cross-fleet risk fan-out** that reconciles to the single-node aggregate to 1e-12.

Cross-platform GPU acceleration

For the embarrassingly-parallel workloads — large strike/tenor batches, Monte-Carlo path-dependent exotics, scenario grids, pathwise Greeks — Celnet drives a pricing-backend abstraction over the cross-platform GPU stack (Metal, Vulkan, DX12), backed by a high-precision CPU oracle and a CPU-SIMD fallback. A counter-based RNG is bit-identical between CPU and GPU, and every GPU result is reconciled against the f64 CPU oracle node-by-node within a *derived* (never fitted) bound, so acceleration never costs determinism. Four GPU kernels ship:

- **Batch closed-form revaluation** — one dispatch prices an $O(10^4-10^6)$ -point (pair, tenor, strike) grid of Garman-Kohlhagen vanillas analytically, one thread per instrument; no RNG, no Monte-Carlo noise, only f32 round-off vs the f64 oracle (measured max rel-err $\sim 3.1e-7$, inside the per-node bound).
- **Multi-step QMC path kernel** — evolves a large batch of multi-step Brownian-bridge paths under *shared* Sobol draws and prices a path-dependent arithmetic-average Asian; the Sobol integers are bit-identical on both sides.
- **Pathwise / likelihood-ratio Greeks** — one dispatch produces per-path delta/vega estimators (pathwise for the smooth call, likelihood-ratio for the discontinuous digital), reconciled three-way GPU-f32 $\approx$ CPU-f64 $\approx$ analytic within MC error.
- **Scenario grid** — a whole spot $\times$ vol shock grid under common random numbers in one dispatch.

The measured dispatch-amortization curve runs **0.68 $\times$ @4k $\rightarrow$ 53 $\times$ @1M paths** as the device saturates. GPU correctness is proven today (GPU-f32 == CPU-f64 within the derived bound, both == the closed-form / golden chain within QMC error) with host-local ratios reproducible on any silicon; the CUDA/NVIDIA absolute GPU throughput headline, the ≤ 50 ms exotic target, and the workload absolute figures are delivered through the CUDA backend on NVIDIA hardware — the f64-correct kernels and reconcile harness are already in place, so the program lights up the absolute numbers on the target accelerator.

Horizontal scale-out fabric

When one node is not the whole firm, Celnet fans out. An intra-fleet highest-random-weight (rendezvous) partition map assigns each currency pair / instrument deterministically to a node, so the trading universe shards cleanly with minimal reshuffling when nodes join or leave — every node agrees on ownership without a coordinator on the hot path. The currency pair is the primary partition key, so *every tenor and strike of a pair is co-resident on one shard* and surface-shaped roll-up never crosses a shard. Pricing and risk for a partition live where that partition lives; clients are routed to the owning node, and capacity grows by adding nodes rather than widening a central server.

Fan-out to counterparties is governed for both fast and slow consumers. The SPMC broadcast ring shares per-pair compute (the LMAX-Disruptor multi-consumer shape: each consumer owns its cursor and re-reads the same slots), and on top of it a conflating egress governor collapses superseded updates per subscriber, paces with a token

bucket, and counts what it drops; the multiplex streaming session carries many subscriptions over one bidirectional channel with per-subscription sequencing and gap-driven resync.

SCALING LAYER	WHAT IT PROVIDES	HOW A DESK USES IT
Node-local hot substrate	Pinned, zero-allocation, lock-free in-core pricing of an IB-sized book at nanosecond scale (price + 13 Greeks)	One node prices a whole book and streams it; add nodes for capacity, not to relieve the maths
GPU backend	Cross-platform (Metal/Vulkan/DX12) batch closed-form, QMC path, pathwise/LR Greeks, scenario kernels; CPU oracle + SIMD fallback; deterministic CPU↔GPU	Offload large batch / Monte-Carlo / scenario work; reproducible on any silicon
Raft-replicated event log	Leader election (Pre-Vote), log-matching, conflicting-tail truncation, snapshot compaction + snapshot reseed; bit-identical replay	Replicate priced-book state across nodes with proven cross-node bit-identity and hot-standby takeover
Broadcast fan-out ring	One producer per pair → N consumers; zero-alloc lock-free publish; bounded conflation with exact skip-accounting	Fan one pair's tick to 100s–1000s of subscriber sessions without per-subscriber recompute
Cross-fleet risk fan-out	Rendezvous-hashed partitioning + additive merge / non-additive re-gather; fan-out == single-node	Compute firm risk by fanning out across shards and reconciling to the single-node aggregate
Conflating egress governor	Per-client conflation, token-bucket pacing, counted drops	Stream to slow and fast counterparties from one source without head-of-line stalls

Distributed correctness: Raft-replicated log + cross-fleet risk

Sharding alone is not a firm-grade distributed system; the state has to stay *correct* across nodes through failures. The load-bearing safety invariant of the replicated log — proven by dedicated capability checks for election, snapshot and compaction — is that any two nodes that have committed index i hold a byte-identical log prefix and bit-identical applied book state; at most one leader exists per term, and a stale leader's divergent tail is reconciled by truncation. Hot-standby takeover loses zero committed entries.

Firm-level risk fans out and back in over the same partition fabric. Each logical shard rolls up its own facts in its own cube; the firm reducer then combines **additive** measures (per-ccy net Greeks, the tenor×delta vega ladder) by associative/commutative merge — exactly the single-node sum, bit-equal under the pinned deterministic shard order — and re-derives **non-additive** measures (VaR / ES, FRTB-SbM curvature, correlation-weighted vega) by *gathering the union of every shard's constituent positions and re-deriving once over the whole set*. Summing shard VaRs would be wrong — VaR is a non-linear functional of the joint P&L distribution, so offsetting positions on different shards diversify (sub-additivity) and adding standalone tails double-counts. Because the firm reducer sees the identical constituent multiset the single-node cube would, the result is exact by construction: the capability checks reconcile **fan-out == single-node firm aggregate to 1e-12** across the full net-Greek set and the vega ladder, and the non-additive VaR/ES/curvature to 1e-12 relative.

One scaling model, no rewrite

The crucial property is that these layers compose without changing the contract. The identical unversioned API, the identical hot-core pricing, and bit-identical values hold whether Celnet runs as a single self-contained node on a trader's desk or as a sharded, GPU-accelerated, Raft-replicated, fanned-out fleet inside the firm's estate. A deployment moves along the scale-out curve by swapping adapters and adding nodes — never by re-implementing the platform — and zero-cost observability (coordinated-omission-aware tail-percentile latency histograms on a bounded drop-on-full telemetry ring) plus regression-gated benchmarks keep that scaling measured and verified as the fleet grows.

3. Quantitative Coverage, the Volatility Surface & Integrated Risk

The analytics surface end to end — vanilla and the full Greek set, the five-family surface engine, the full exotic catalogue, the validation regime, and how one source of truth rolls up into firm-wide risk and capital & XVA. The same methods generalize across asset classes in §4.

Celnet is not a thin challenger closing gaps — it is a functionally complete, evidence-backed superset of the quant catalogue a derivatives desk stitches together today, proven by a runnable parity matrix against independent oracles and reachable identically from five clients. Every price, every Greek, every smile and every exotic is computed by the **same in-core engine** that powers the live pricing edge, so the number a trader sees in the ticket, the number that streams over the wire, and the number a quant pulls into a spreadsheet are one and the same. This section maps the analytics surface end to end — from a single vanilla through the full Greek set, the five-model smile engine, the full exotic and structured catalogue, and the LSV booking engine — and then shows how that single source of truth is rolled up into firm-wide risk, regulatory capital and counterparty valuation adjustments. Each method is the state-of-the-art technique for its regime; the methods catalogue and its provenance are in §5.

The Celnet quant coverage map — a single in-core library spanning Garman–Kohlhagen vanilla pricing, the full FX desk Greek set, the four delta conventions, the five-model smile/surface engine, and the full exotic & structured catalogue (vanilla-family · path-dependent · structured/MC · American–Bermudan · correlated basket · LSV booking), validated by ~26 capability checks against independent oracles plus frozen QuantLib golden tables, with the Open Quant SDK as the extension seam to bespoke products.

Vanilla pricing and the full FX desk Greek set in one pass

Vanilla European options are priced with the Garman–Kohlhagen model, struck off the outright forward and discounted with **separate domestic and foreign discount factors** — the correct FX construction, not an equity model bent into shape. From that single valuation Celnet returns **price plus the full FX desk Greek set in one pass**: spot-delta and forward-delta, gamma, vega, theta, both domestic and foreign rhos, and the full second- and third-order book — vanna, volga, charm, speed, zomma and color. A desk gets every sensitivity it risks against from a single call, with no second pricing round-trip to chase a cross-Greek.

Every one of these sensitivities is **cross-validated by finite differences** against the analytic closed form, so the analytic Greeks a desk trades on are continuously checked against an independent numerical bump of the same model — correctness is a property of the build, not a hope. The hot core that produces price plus 13 Greeks is the pinned zero-alloc loop benchmarked at nanosecond scale.

GREEK FAMILY	SENSITIVITIES DELIVERED IN THE SINGLE PASS
Value	Price (off the outright forward, dual discount factors)
First order	Spot-delta, forward-delta, vega, theta, rho-domestic, rho-foreign

GREEK FAMILY	SENSITIVITIES DELIVERED IN THE SINGLE PASS
Second order	Gamma, vanna, volga, charm
Third order	Speed, zomma, color

Delta conventions and the branch-aware strike↔delta solver

FX desks quote in delta, not strike, and the mapping between them depends on convention. Celnet implements all **four delta conventions** — spot and forward, each in unadjusted and premium-adjusted form — and a **branch-aware strike↔delta solver** that inverts the relationship robustly. Crucially, the solver is aware of the **premium-adjusted call-delta maximum**: where the premium-adjusted delta function turns over and a naive root-finder would pick the wrong branch, Celnet selects the correct one. Both at-the-money rules are built in and applied **sign-correctly per convention** — at-the-money-forward (ATMF) and the delta-neutral straddle (DNS). A trader can pin a wing by 25-delta or 10-delta, anchor the smile at ATMF or DNS, and trust that the strike Celnet returns is the strike the market means.

Adjoint Greeks at portfolio scale

For risk at firm scale the cost of sensitivities matters as much as their correctness. Celnet computes Greeks by genuine **reverse-mode adjoint algorithmic differentiation (AAD)** over the pricing graph: a single reverse sweep yields the full first-order block plus the second-order gamma/vanna/volga, turning an $O(\text{factors})$ bump-and-revalue into an $O(1)$ -in-factors pass. The closed-form analytic Greeks are retained as the validation oracle, gated equal to $\sim 1e-12$, so the fast path is provably the exact path. This is what lets the risk cube re-derive a whole book's sensitivities in one sweep per position rather than one per shocked factor.

3.1 The Volatility Surface Engine — Five Smile Families, Arbitrage-Free by Construction

How Celnat marks vol: five selectable smile models, broker-strangle calibration, no-arbitrage gates, and a Dupire local-vol surface that feeds the exotic engines.

Celnat marks volatility with a full smile-and-surface engine rather than a single fixed parameterisation. **Five smile models** are available — **Vanna-Volga**, **SABR**, **raw-SVI**, **SSVI** and **eSSVI** (the extended, maturity-dependent- ρ surface) — and a **smile-model selector** lets a desk mark or recalibrate a smile under any of them and compare. SSVI is byte-recovered as the constant- ρ special case of eSSVI, so the extension is additive, not a fork. Market quotes enter the engine the way they are actually traded: a **broker-strangle** → **smile-strangle fixed-point calibration** resolves the market (broker) strangle into a consistent smile strangle, so the calibrated smile reproduces the prices a desk was shown.

No-arbitrage is enforced, not assumed. **Arbitrage gates** check butterfly density non-negativity (no negative implied densities, via pointwise Breeden–Litzenberger re-pricing), vertical-spread monotonicity, and calendar total-variance monotonicity across tenors. The eSSVI surface carries the closed-form static no-arbitrage conditions (per-slice butterfly plus consecutive-slice calendar). The term structure is assembled **arbitrage-free, interpolated in total variance**, so volatility between pillar tenors is consistent and free of calendar arbitrage by construction. For the exotic and LSV engines, the smile is converted to a **Dupire local-volatility** surface.

The surface pipeline — broker (market) strangle quotes enter a smile-strangle fixed-point calibration, pass butterfly / vertical / calendar arbitrage gates, are marked under a selectable smile model (VV / SABR / SVI / SSVI / eSSVI), and are woven into an arbitrage-free term structure interpolated in total variance and extracted to a Dupire local-volatility surface for the exotic engines.

STAGE	WHAT THE ENGINE DOES
Quote intake	Broker (market) strangle → smile-strangle fixed-point calibration
Smile models	Vanna-Volga, SABR, raw-SVI, SSVI, eSSVI — chosen via the model selector
Arbitrage gates	Butterfly density non-negative, vertical monotonic, calendar total-variance monotonic
Term structure	Arbitrage-free, interpolated in total variance
Local vol	Dupire local-volatility extraction feeding the exotic/PDE/LSV engines
Marking	Mark / recalibrate / publish under any model with surface versioning

The marking workflow is exercised live from the GUI's Surface workspace — a smile chart alongside the ATM / 25RR / 25BF / 10RR / 10BF marking grid, the arbitrage-free gate, the broker-calibrated smile, an explicit surface version with reset and publish controls, and five model chips including eSSVI.

The Surface workspace in the live GUI — a smile chart, the ATM / 25RR / 25BF / 10RR / 10BF marking grid, the arbitrage-free gate, the broker-calibrated smile, the explicit surface version, the five model chips (VV / SABR / SVI / SSVI / eSSVI), and the Reset / Publish controls that deposit a fresh marked surface.

3.2 The Full Exotic & Structured Catalogue

Every exotic and structured product as a real shipped engine, organised by the numerical regime that sets its validation bar — the breadth that meets the deep-catalogue platforms, extensible to bespoke IP via the Open Quant SDK.

On top of vanilla and the smile engine, Celnet ships the **full FX exotic and structured catalogue** — not a first-generation subset. Every product below is a real, shipped engine, exposed on the one wire as an arm of the unified instrument vocabulary, and gated by an independent capability check against an independent oracle. The catalogue is organised by the numerical regime each product class genuinely lives in — which is exactly what determines its validation bar.

(a) First-generation vanilla-family exotics — method triangulation

Digitals, one-touch and no-touch, double-no-touch and double-touch, single and double barriers (knock-in and knock-out), and window barriers. What sets this band apart is **method triangulation**: each product is reachable by multiple independent methods, cross-validated against one another so PDE, Monte-Carlo and analytic agree before a price is trusted.

- **Analytic** — closed-form barrier valuation via the reflection-principle construction.
- **Crank-Nicolson PDE** — a finite-difference solver with a **Rannacher start-up** that damps the oscillations barriers and digital payoffs would otherwise induce near the boundary.
- **Counter-based Monte-Carlo** — a Philox simulation path, bit-reproducible across runs and across CPU and GPU.
- **Survival-weighted Vanna-Volga overlay** — a market overlay that re-introduces smile risk into the touch / barrier price, weighted by survival probability.

For this band, **PDE ≈ MC ≈ analytic** is a continuously enforced invariant, and every product is validated against the **frozen QuantLib golden tables**.

PRODUCT	ANALYTIC	CRANK-NICOLSON / RANNACHER PDE	PHILOX MONTE-CARLO	SURVIVAL-WEIGHTED VV OVERLAY
Digitals	✓	✓	✓	✓
One-touch / No-touch	✓	✓	✓	✓
Double-no-touch / Double-touch	✓	✓	✓	✓
Single barriers (KI / KO)	✓	✓	✓	✓
Double barriers (KI / KO)	✓	✓	✓	✓

PRODUCT	ANALYTIC	CRANK-NICOLSON / RANNACHER PDE	PHILOX MONTE- CARLO	SURVIVAL-WEIGHTED VV OVERLAY
Window barriers	—	✓	✓	✓

(b) Closed-form structured products

Priced by genuine closed-form or model-free static replication, then gated at closed-form precision against an independent recomputation:

- **Variance swap** — fair variance strike by **log-contract $1/K^2$ static replication** (Demeterfi–Kamal–Zou / Carr–Madan) over the smile-consistent OTM forward-option strip, with an adaptive wing extended to a relative-convergence floor.
- **Volatility swap** — the **Carr–Lee convexity (Jensen) adjustment**, giving a fair-vol strike strictly below $\sqrt{k_{\text{var}}}$ for any non-degenerate smile.
- **Arithmetic Asian** — the arithmetic average is not lognormal, so there is no exact form; Celnat ships the two market-standard fast analytic estimators, each labelled at its true accuracy: **Turnbull–Wakeman** two-moment lognormal matching and **Curran** geometric-conditioning, handling the FX carry and the seasoned (in-progress-average) case.
- **Forward-start vanilla & cliquet** — the exact **Rubinstein (1990)** FX dual-carry strike-reset closed form; a plain (uncapped) cliquet is the exact sum of its forward-start legs, while a locally-capped/floored cliquet falls to Monte-Carlo.
- **Quanto** — exact closed form via the quanto drift adjustment, with the Monte-Carlo engine as a cross-check.
- **Lookback** — fixed- and floating-strike closed forms (Goldman–Sosin–Gatto / Conze–Viswanathan) for continuous monitoring, with a Brownian-bridge extremum-correction Monte-Carlo that converges to the continuous-monitoring oracle.

(c) Monte-Carlo path-dependents

Products whose payoff structure has no closed form, priced on the counter-based, antithetic, bit-reproducible Monte-Carlo engine over the scrambled-Sobol / Brownian-bridge quasi-random stack. Each carries a Monte-Carlo price standard error on the wire as a precise statement of estimator precision:

- **TARF** — target-redemption forward: a strip of periodic fixings with a cumulative target that knocks the structure out, gearing on the adverse side, and the explicit full-gain vs capped-gain gap-risk premium.
- **Accumulator / decumulator** — periodic accumulation at a discounted pivot with a knock-out barrier and adverse-side gearing, in discrete and Brownian-bridge-continuous monitoring conventions.
- **Correlated multi-asset basket / best-of / worst-of** — weighted basket, best-of-N and worst-of-N (rainbow) calls and puts over a portfolio of pairs, priced by a **Cholesky-correlated** multi-asset GBM Monte-Carlo over the QMC stack; a non-positive-definite correlation matrix is rejected explicitly rather than silently regularised.

(d) American / Bermudan early exercise

Physically-settled FX options trade American-style, and Celnat prices them **two independent ways and cross-validates them**: a **projected-SOR (PSOR) free-boundary Crank–Nicolson finite-difference** solver that treats early exercise as a linear-complementarity problem and discovers the exercise boundary without tracking it, and a **Longstaff–Schwartz regression Monte-Carlo (LSM)** over Sobol/Brownian-bridge paths. The Bermudan case applies the exercise projection only at permitted dates; the LSM estimator carries a Monte-Carlo price standard error.

(e) LSV booking engine + standalone Heston

The booking-grade second-generation layer:

- **Local-Stochastic-Volatility booking model** — a Heston variance backbone with a Dupire **leverage** surface calibrated by the **interacting-particle method** (the McKean–Vlasov leverage identity, calibrated forward-in-time, non-parametrically via a Nadaraya–Watson kernel estimate of the conditional expected variance). It prices on **two independent engines** — a 2-D Hundsdorfer–Verwer **ADI PDE** on the (spot, variance) grid and the antithetic QE-stepper Monte-Carlo — anchored by three asserted guarantees: it reprices the arbitrage-free vanilla surface (PDE), $\text{ADI-PDE} \approx \text{MC}$ on a window-barrier payoff, and the pure-local-vol limit recovers the Dupire price. Selectable on the wire via the pricing-model directive (default analytic vs local-stochastic-vol).
- **Standalone Heston** — European vanilla via **two genuinely independent Fourier transforms** of the same branch-cut-free (Cui–del Baño Rollin–Germano) characteristic function: **Carr–Madan** damped-integral Gauss–Legendre quadrature and the **Fang–Oosterlee COS** method, agreeing tightly over the $\leq 3y$ FX grid, and gated against a frozen **QuantLib golden table**. No external FFT dependency — single-strike pricing controls accuracy directly.

Sobol quasi-Monte-Carlo variance reduction

The Monte-Carlo path-dependents ride a complete **randomized quasi-Monte-Carlo (RQMC)** stack: a gray-code **Joe–Kuo Sobol'** generator with an embedded open-source direction-number table, an **Owen-style nested digital scramble** for unbiased replications, principal-bisection **Brownian-bridge** path construction loading the dominant variance onto the best-distributed dimensions, and a full-precision inverse-normal CDF. The variance reduction is **measured, not asserted** (gated $\geq 3\times$): on exact-value targets the QMC RMSE is roughly **38× smaller for a geometric-average Asian** and **~88× smaller for a European** than a fair plain pseudo-random estimator at the same path budget.

~26 capability checks · 5 smile models · 18+ product families · QuantLib 1.42.1 golden tables · QMC RMSE up to ~88× tighter

3.3 Validation Regime — the Right Oracle for Each Product Class

The dates, conventions and independent oracles underneath every price — each product class gated against the strongest standard it admits (the full assurance estate is in §8).

Pricing is only as good as the dates, conventions and oracles underneath it. Celnet runs a **dual-calendar date engine** — resolving spot, expiry, delivery and roll across the two settlement calendars an FX pair actually depends on, including ON/TN/SN, IMM and broken dates — and a **per-(currency-pair, tenor) convention registry** carrying the right delta convention, ATM rule, day-count and premium treatment per instrument, validated against published EMTA/ISDA tables across a 19-pair universe.

The validation method is stated per product class rather than blanket-claimed — each band is gated against the oracle that genuinely bounds it, and Monte-Carlo products report estimator precision explicitly:

PRODUCT CLASS	VALIDATION ORACLE	BAR
Vanilla, digitals, both touch styles, all 8 barriers, window barrier	Frozen QuantLib 1.42.1 golden tables + analytic/PDE cross-validation	~1e-9/1e-10 (closed-form), ~1e-6 (series-gated double-barrier)
Standalone Heston	Frozen QuantLib golden table + Carr–Madan-vs-COS internal agreement	tight relative + absolute band
Variance/vol swap, forward-start/cliquet, lookback (continuous), quanto	Independent closed-form recomputation / closed-form limits	~1e-6 to ~1e-10
Asian (Turnbull–Wakeman, Curran)	Closed-form degenerate limits + independent MC within reported stderr	TW at its true ~1.5% approximation band; Curran within MC stderr
eSSVI surface	Pointwise Breeden–Litzenberger density + calendar-monotone re-pricing; SSVI byte-recovery	density ≥ 0, ≤1e-9 self-reprice
TARF, accumulator, basket/best-of/worst-of, American/Bermudan (LSM)	Independent code-disjoint MC / closed-form limits	within reported price std-error band
LSV booking model	Vanilla-surface reprice + ADI-PDE ≈ MC + pure-local-vol Dupire limit	tolerance-gated, asserted in-suite

The whole library is re-checked on every build by **~26 capability checks against genuinely independent oracles** plus the frozen golden tables — never plausibility, always agreement with an independent computation. The discipline is sharpened by a real defense: a circular self-oracle defect in the FRTB low-correlation floor was caught and the constant re-derived against the published BCBS source.

Extensible to bespoke products via the Open Quant SDK

The catalogue above is the **shipped core**, already matching the structured/path-dependent breadth the deep-catalogue platforms charge for — not the ceiling. The **Open Quant SDK** exposes the same pricing-model, smile-model and calibration seams the core uses, so a desk can extend Celnnet to **house and bespoke payoffs** and run them **inside the same engine** — sandboxed (the native or the deterministic WebAssembly sandbox), bit-reproducible by replay, and hot-loadable, without forking Celnnet. Structures like TARFs, quantos and baskets are **already shipped core**, so the SDK is for genuinely novel IP rather than filling catalogue gaps.

The Ticket workspace is where this breadth becomes a workflow: a structure selector spanning the full catalogue (vanilla strategies, the first-generation exotics, Asians, forward-start/cliquet, quanto, lookback, TARF, accumulator, window barriers, American/Bermudan, and correlated baskets/best-of/worst-of), the notional and tenor strip, the legs and strikes, a zero-cost Solve, two-way BID / MID / OFFER, the active conventions shown on the face of the ticket, and direct hand-off to Request-quote, Stream or Add-to-risk.

The Ticket workspace — a full-catalogue structure selector (vanilla strategies through American/Bermudan and correlated baskets), notional, the tenor strip, legs with strikes, a zero-cost Solve, two-way BID / MID / OFFER, the active conventions shown on the face, and direct hand-off to Request-quote / Stream / Add-to-risk.

3.4 Integrated Risk — One Fact Cube, Two Altitudes

One position-fact cube seen at two zooms — a position's behaviour under shocks and the firm's whole netted book — the algebra that generalizes to every asset class in §4.4.

Celnet treats risk as a single, integrated capability that spans two altitudes at once: the *micro* view of an individual position's behaviour under shocks, and the *macro* view of the firm's entire option book netted, rolled up, and policed against limits. Both are two zooms of one underlying truth — a position-fact cube — so a trader and a board-level risk officer are always looking at the same numbers, never a reconciliation of two systems.

Risk architecture — a versioned marked-surface registry feeds book-shaped scenario risk (the desk's working view) and a convention-canonicalized OLAP position-fact cube (the firm-wide view) that aggregates vanilla **and exotic** legs. Normalize → cube → limits ∥ entitlements: one fact model, drilled at every altitude, extended to FRTB-SA regulatory capital and internal XVA.

Book-shaped scenario risk

Risk at the desk is computed by real repricing, not by linear extrapolation from a single point. Every scenario number is produced by re-running the same nanosecond-scale pricing core that quotes live, so the risk surface is internally consistent with the price.

MEASURE	WHAT IT GIVES THE DESK
Two-axis scenario grid	A shock matrix — typically spot × vol — where every cell is a genuine reprice. Axes are swappable, so the same grid renders P&L, delta, or vega exposure.
Bucketed vega ladder	Vega decomposed per (tenor, delta) pillar, so volatility risk is read where the desk actually marks it — short-dated vs long-dated, wings vs ATM.
Cross-gamma stencil	A two-dimensional curvature stencil capturing how delta moves as spot <i>and</i> vol move together, not just along a single axis.
Theta-roll	Time decay projected over forward horizons, rolling expiries node by node so the desk sees the carry of the book as the clock advances.

Underpinning all of it is a **versioned marked-surface registry**. A calibrated smile is deposited under a fresh surface version and becomes the official mark; pricing and risk then *pin* against a named version. The registry holds the distinction between the official mark and the live market explicitly — official-vs-live is a first-class concept, not an overwrite. Asking for an unknown surface version is rejected outright rather than silently falling back to live data, so a risk run is always reproducible against exactly the surface it claims to use.

The Risk workspace (Cmd-4) — a spot × vol reprice grid with swappable axes and P&L / delta / vega tabs, the vega ladder bucketed per tenor × delta, plus the cross-gamma stencil and theta-roll, every cell a genuine reprice on the live pricing core.

The firm-wide position-fact cube

Above the desk view sits an OLAP **position-fact cube** that gives the firm one canonical, queryable picture of all option risk. Positions arrive in many conventions and numeraires; Celnet canonicalizes them — convention-normalized and netted into a common numeraire — before anything is aggregated, so figures across pairs, books, and entities are genuinely additive where they should be. The cube is an immutable fact table over eight dimensions: Trader, Book, Desk, CurrencyPair, Location, Entity, ValueDate and Session.

Roll-up respects the mathematics of each measure. **Additive** measures (notional, delta, vega) accumulate incrementally up the hierarchy — fast, and exact. **Non-additive** measures (VaR, expected shortfall, curvature) are *re-derived per node* rather than summed, because a portfolio's tail risk is not the sum of its parts. The cube knows the difference and applies the correct treatment automatically at every level.

On top of the cube sits a **cascading limit tree** — board → entity → desk → book → trader — with both pre-trade and post-trade checks, so a candidate trade is tested against every limit it would touch before it is done, and the live book is policed continuously after. Finally, aggregation is **entitlement-aware**: the server prunes the cube to what a given viewer is allowed to see *before* it aggregates — pre-aggregation pruning, not after-the-fact redaction.

One cube, two views — Book ↔ Risk drill

Because the desk's book view and the scenario-risk view are projections of the same fact cube, a trader can move between altitudes in a single gesture. The **Book** workspace presents net P&L, Vega, Gamma, and Theta as headline cards with a per-pair breakdown and an aggregate vega ladder; selecting any aggregated book row **drills straight into that position's scenario risk** in the Risk workspace — same numbers, deeper zoom, no context switch and no separate tool.

The Book workspace (Cmd-5) — net P&L / Vega / Gamma / Theta cards, a per-pair breakdown that drills straight into a single position's scenario risk, and the aggregate vega ladder rolled up from the server-side position-fact cube.

Exotic legs aggregate as first-class citizens

The cube is not a vanilla-only ledger. A booked exotic — a barrier, a digital — earns a genuine seat in every roll-up, contributing its **real** sensitivities rather than a vanilla proxy or a silent zero that would understate the firm's risk. An exotic leg supplies its full canonical Greek set and premium line so it sums into the net-Greeks and vega-ladder roll-up exactly like a vanilla leaf, and it re-prices the *real exotic payoff* under each scenario for VaR/ES and supplies its own up/down curvature legs for FRTB — so the firm tail and curvature reflect a knock-out's gamma sign-flip near the barrier or a digital's pin risk, never a smoothed-over vanilla approximation.

Digitals use closed-form digital Greeks; barriers use a central finite-difference of the closed-form Reiner-Rubinstein price (deterministic and bit-reproducible). The deterministic, closed-form members — single barrier and European digital — re-price exactly under shocks with no estimator noise, so they slot into the cube's machine-exact VaR/curvature path; MC-priced exotics extend the same seam through a finite-difference of a common-random-number MC price that carries an explicit Monte-Carlo standard-error band.

Crucially, this exotic-leg aggregation is preserved end-to-end through the distributed fan-out: a firm book *including* exotic legs, partitioned across a rendezvous-hashed fleet and re-aggregated through the cross-shard algebra, reconciles to the single-node firm aggregate — additive Greeks to $\sim 1e-12$ and the re-gathered non-additive VaR/ES and curvature to the same bit-level tolerance. The exotic seat survives sharding; the firm view is the same number whether computed on one node or fanned out across the fleet.

3.5 Regulatory Capital & Counterparty Valuation Adjustments

FRTB-SA standardised capital and an internal CVA/DVA/FVA lens computed on the same fact cube — the risk-class-agnostic machine that lights up cross-asset capital in §4.4.

FRTB-SA (Standardised Approach)

On top of the position-fact cube, Celnet computes market-risk capital under the Basel **FRTB Standardised Approach** (BCBS MAR21/22/23) — the full Sensitivities-based Method (SbM), not a stand-in. Capital is built bottom-up from net sensitivities the cube already holds:

STAGE	METHOD (BCBS REFERENCE)
Weighted sensitivity	$WS_k = RW_k \cdot s_k$ — each net delta/vega scaled by its prescribed risk weight (MAR21.3).
Within bucket K_b	$K_b = \sqrt{(\max(0, \sum WS_k^2 + \sum_{k \neq l} \rho_{\{kl\}} WS_k WS_l))}$, with the MAR21.4(3) non-negativity floor under stressed correlation.
Across buckets	$K = \sqrt{(\sum K_b^2 + \sum_{b \neq c} \gamma_{\{bc\}} S_b S_c)}$ with signed bucket sums S_b , and the MAR21.6 alternative $S_b = \max(\min(\sum WS_k, K_b), -K_b)$ when the cross-bucket radicand goes negative.
Three correlation scenarios	The whole charge is recomputed under HIGH ($\rho, \gamma \rightarrow \min(1, 1.25 \cdot \rho)$), MEDIUM (as prescribed) and LOW ($\rho, \gamma \rightarrow \max(2 \cdot \rho - 1, 0.75 \cdot \rho)$) correlations, and the capital is the maximum of the three — the defining SbM property (MAR21.6(1)).
Curvature	Per-bucket $K_b = \max(K_b^+, K_b^-)$ from the up/down reprice net of delta, with the ρ^2/γ^2 curvature correlations and the chosen direction's CVR sign carried into the cross-bucket sum (MAR21.5.2).
RRAO	A flat add-on — 1.0% of gross notional on exotic-underlying instruments plus 0.1% on instruments carrying other residual risk (gap/digital, correlation, behavioural). Barriers, digitals, one-touches and TARFs are textbook RRAO (MAR23.4/.5).
DRC	For a pure deliverable-FX book the Default Risk Charge is a documented, cited zero (MAR22) — there is no issuer jump-to-default in deliverable FX.

The quadratic-form kernel $\sqrt{(\sum WS^2 + \sum \sum \rho WS WS)}$ is factored once and reused for delta, vega and the cross-bucket step, so there is no duplicated formula and no opportunity for the three sites to drift. The **0.75ρ** LOW-correlation floor — material for FX where $\gamma = 0.6$ must decorrelate to **0.45**, not **0.2** — is pinned to **hand-computed BCBS constants** in a dedicated test, after a prior wave caught a circular-oracle defect (a longhand check that had independently re-derived the *same* wrong floor). The whole SbM machinery is gated against a genuinely independent longhand oracle — every formula re-written from raw weighted sensitivities using only primitive double-precision arithmetic, never calling back into the production code — agreeing to ~1e-10, with structural identities (a perfectly-hedged bucket gives $K_b = 0$; RRAO equals the exact hand-summed sum of $|notional| \cdot weight$).

XVA — CVA / DVA / FVA

Celnet computes the counterparty-risk valuation adjustments — **CVA** (credit), **DVA** (debit/own-credit) and **FVA** (funding) — over a netting set of vanilla FX options. The pipeline is end-to-end and exact: an exposure simulation evolves spot under risk-neutral GBM on an exposure-date grid, driven by low-discrepancy **Sobol** normals (far lower exposure-profile variance than plain pseudo-random MC at the same path budget), reprices and **nets within the set** at each grid date, and reduces across paths to the expected positive/negative exposure profiles; a piecewise-constant hazard-rate curve gives risk-neutral survival; and the discrete Basel/ISDA aggregation forms unilateral CVA over discounted expected positive exposure and the survival-decrement, the symmetric DVA over own-survival and negative exposure, and the funding adjustment FVA over the net expected exposure on the joint-survival measure. The arithmetic is validated against a hand-derived closed-form CVA in exact limits — agreeing to $\sim 1e-9$ with every intermediate constant **hand-pinned to offline-computed published values**, monotone in hazard and LGD, zero at zero default probability, with DVA/FVA sign symmetry.

Method confidence and the delivery program. Monte-Carlo-priced products (TARF, accumulator, discrete lookback, basket/best-of/worst-of, American via LSM, capped/floored cliquet) carry a **price standard error** as a first-class numerical-confidence band, while the analytic/PDE/golden-gated set holds to the machine-precision bar — two complementary rigour levels, each appropriate to its product class. The XVA engine is a full CVA/DVA/FVA valuation engine over netting sets on a risk-neutral GBM exposure model; Celnet's implementation program extends it onto live CSAs / collateral, wrong-way risk and the live credit/funding-curve estate on the same exposure pipeline. The cross-fleet risk fan-out algebra reconciles fan-out == single-node to $\sim 1e-12$, with physical cross-node transport and cross-host wire latency delivered on the built HRW seam. GPU correctness and ratios are proven against the f64 CPU oracle today, with the CUDA/NVIDIA absolute throughput headline delivered through the CUDA backend on NVIDIA hardware.

4. Multi-Asset Architecture & Cross-Asset Capability

The platform's second axis of leadership — one carry-producing-market architecture that turns a new asset class into a new carry builder, the asset-class superset, the linear book and multi-dealer RFQ, and one firm book across every class. The methods generalize from \$3; the architectural leap is detailed in \$5.

Celnet already exceeds the single-dealer FX-options incumbents on *product* breadth — eighteen on-wire product families, five smile models, LSV and standalone Heston, GPU, FRTB and XVA, all behind one unversioned contract. The platform's next axis of leadership is *asset-class* breadth, and it is built on a deliberate architectural decision rather than a pile of per-class special cases. That decision is the **carry-producing-market architecture**: a single contract whose pricing engines never know what asset class they are pricing, so a new asset class is a new *carry builder* — not a new pricer, not a contract fork, not a branch threaded through every payoff. FX is the live, gate-validated asset class today; precious metals, digital-assets/crypto, equity/index, commodity and rates are the cross-asset capability that the same architecture reaches, each validated against an independent oracle and surfaced identically across all five clients.

The carry architecture at the centre, asset classes radiating out. Three orthogonal layers — an underlying-identity layer, a carry forward/discount producer, and asset-class-agnostic payoff engines — mean a new asset class is a new carry builder with *zero payoff-engine edits*. FX is live and gate-validated; precious metals, crypto, equity, commodity and rates are the cross-asset capability on the same seam, with the linear book, structured Pivot, multi-dealer RFQ and a cross-asset risk cube built on top.

4.1 The cross-asset architecture: identity, carry, agnostic payoff

One forward/discount producer, many carry assemblies — the payoff side never branches on asset class. Every class maps its native inputs into the single generalized-BSM forward $F = S \cdot e^{(b \cdot t)}$ and discount $df = e^{(-r \cdot t)}$: FX ($b = r_{\text{dom}} - r_{\text{for}}$, reproduced bit-for-bit), equity ($b = r - q - \text{repo}$), commodity ($b = r - \text{convenience}$), the listed-future Black-76 case ($b = 0$), crypto ($b = r - \text{funding}$) and metals ($b = r - \text{lease}$) differ only in how b is assembled from named factors. The asset-class-agnostic payoff engines consume only the forward, discount and surface; a binding no-workaround rule makes a carry-variant match in payoff code a review blocker, a multi-curve-ready seam slots OIS/CSA discounting behind the same interface, and a typed `INVALID_ARGUMENT` validity matrix replaces any silent fallback.

Celnet began FX-only at three load-bearing seams that ran through every layer: the pricing-inputs and Greeks shape (Garman-Kohlhagen with domestic/foreign rates and two rhos), the wire vocabulary (a currency pair and its spot, vol and two rates), and the pricing-model plugin contract. The decision is to split that one fused FX seam into **three orthogonal layers**, each with a single concern — generalize in place, one clean contract, no version negotiation, FX byte-identical.

LAYER	CONCERN	WHAT IT IS
1 • Identity	Name the asset class — <i>once</i>	An underlying-identity union (FX today; precious metals, digital assets, equity and listed futures as additive growth) carrying asset class plus settlement style (deliverable / cash-at-fixing / inverse-coin) and quote orientation. This is the <i>only</i> place asset class is named anywhere in the platform.
2 • Market	How the underlying drifts & discounts	The carry layer is a <i>producer</i> of the forward and the discount factor — not a variant pricers branch on. FX two-rate, equity (rate, dividend), commodity (rate, convenience) and crypto (rate, funding) all collapse to generalized-BSM $F = S \cdot e^{(b \cdot t)}$, $df = e^{(-r \cdot t)}$; they differ only in how the carry b is assembled from named factors.
3 • Payoff	Price the option — class-blind	Vanilla, barrier, Asian and every other engine consume the forward, the discount and a vol surface <i>only</i> . A new asset class is a new carry / market-state builder with zero payoff-engine edits .

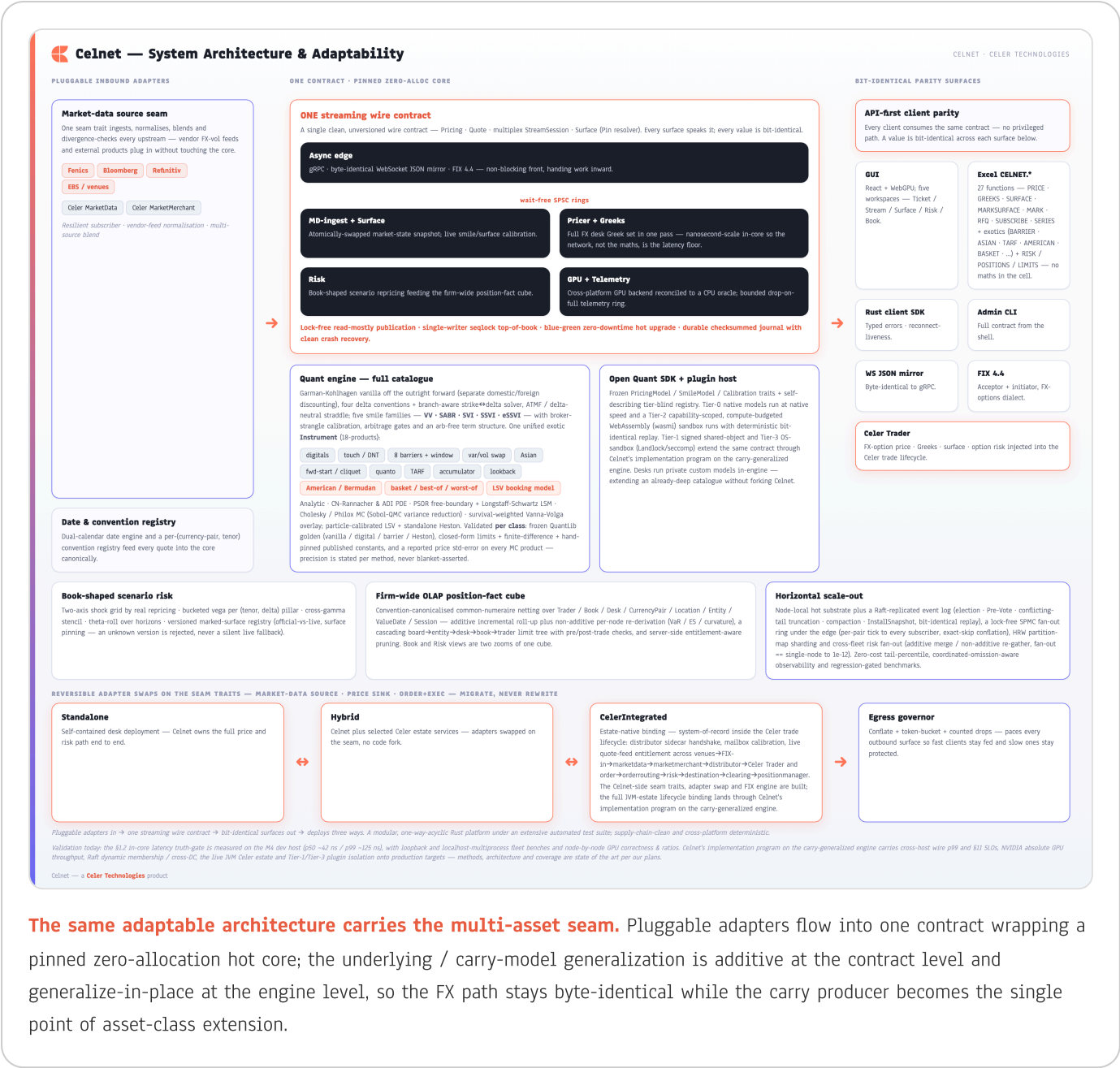
Generalized-BSM with FX as a projection. The unifying mathematics is the generalized Black-Scholes-Merton (r, b) form: an option's forward is $F = S \cdot e^{(b \cdot t)}$ and its discount is $e^{(-r \cdot t)}$. FX is simply the projection where the carry is the two-rate differential — Garman-Kohlhagen falls straight out, and the FX carry reproduces the existing FX forward and discount *bit-for-bit*. That byte-identity invariant is not a nicety; it is the gate the entire architecture is validated against (the golden vectors plus the conformance corpus across all five clients, every value compared on its raw bit representation). Equity is $b = r - q$ (continuous dividend yield, with a repo spread folded in as $b = r - q - \text{repo}$); commodity is $b = r - \text{convenience}$ (and Black-76 on a listed future is the degenerate $b = 0$ priced off the future directly); crypto is $b = r - \text{funding}$. One forward/discount producer, many carry assemblies.

The no-workaround test (binding). A payoff engine that branches on which asset class produced the carry is a *failure* of the abstraction, not a shortcut. Payoff code asks the carry only for the forward and the discount factor and nothing else; any such branch in payoff code is a review blocker. This is what makes "a new asset class is a new carry builder" true rather than aspirational — an NDF on a future crypto or metal underlying reuses the identical linear engine, and a crypto linear vanilla reuses the identical generalized-BSM math the FX path already uses.

Sensitivities: a fixed zero-alloc hot core plus a factor-keyed risk layer. The Greek strip lives in the pinned, allocation-free hot core, so it keeps a fixed flat-data shape — rate sensitivities are a small fixed set (FX domestic/foreign rhos, or a carry discount-rho and carry-rho), never an allocating factor map. The decomposition into FRTB GIRR/equity/commodity buckets happens off the hot path at the *risk layer*, whose roll-up algebra is already factor-generic. The FX relation is exact and recovered by the existing AAD adjoint. This is how "general" reconciles with "zero-alloc" — generality where it costs nothing, fixed flat data where latency is budgeted.

Multi-curve-ready by construction, and intuitive for the FX user. A flat-rate carry is the degenerate single-point curve; an OIS/CSA discounting curve with projection and basis slots in *behind the same* forward/discount interface with zero payoff-engine changes — the proof the seam is in the right place. On the wire the contract gains an underlying selector, a carry-model selector and the generalized rate sensitivities, but every client **defaults to FX**: an existing GUI ticket, a `CELNET.PRICE` cell, or a CLI price for a pair is unchanged — the

asset-class selector is purely additive. There is no silent mis-pricing: a product × underlying validity matrix returns a typed invalid-argument error rather than guessing, so an FX engine handed a non-FX carry errors loudly instead of falling back.



4.2 The asset-class superset

On that architecture sits the asset-class superset — decisively exceeding SynOption's single-asset coverage and matching the heavy front-to-back platforms on span. Every class is delivered on the same carry seam, each reachable through the shared forward/discount interface, each with its own settlement specifics and its own independent oracle. FX carries the full measured truth-gate and golden-table proof today; the other classes are delivered on the identical carry-generalized engine through Celnet's implementation program, each grounded in a validated method and an oracle so it lands as a composable leaf rather than a rewrite. This is comprehensive multi-asset capability on a single coherent contract — one platform spanning the whole asset spectrum where the single-dealer incumbents cover one band.

ASSET CLASS	CARRY / METHOD	SETTLEMENT & CONVENTIONS	STATUS
FX — deliverable + NDF	Garman-Kohlhagen = generalized-BSM with the two-rate carry differential	Deliverable spot/forward; NDF cash-settled in the convertible currency at a named fixing source; 19-pair panel growing toward a >75 superset	Live
Precious metals — XAU/XAG/XPT/XPD + crosses	GK with the metal leg's lease rate in place of a foreign deposit rate ($b = r_{\text{quote}} - \text{lease}$)	Loco-London T+2; metal as base, premium in the quote currency (unadjusted); metal crosses on London n quote-centre; LBMA conventions	Capability
Digital assets / crypto — BTC/ETH/...	Linear: agnostic generalized-BSM with funding carry $b = r - \text{funding}$. Inverse/coin-settled: the $1/S_T$ -denominated payoff via a numeraire change (the new payoff shape)	24x7 continuous trading clock (no weekend/holiday roll); USD(T)-margined vs inverse/coin-margined; UTC exchange-fixed expiry instants; strike/log-moneyness smile	Capability
Equity / index	Generalized-BSM with dividend + repo carry $b = r - q - \text{repo}$; carry-rho = dividend-rho	Cash settlement in the index/quote currency; contract multiplier; strike/moneyness surface	Capability
Commodity — listed-future options	Black-76 on the future ($b = 0$); spot+convenience $b = r - \text{convenience}$ for an option on spot/forward	Futures-settled (cash at the future's settlement); convenience-yield carry	Capability
Rates	Bachelier / normal vol, curve-bucketed rho	Delivered on the same carry interface through Celnet's implementation program — a normal-vol leaf and curve-bucketed rho on the finished seam	Capability

The crypto inverse contract — the precise part. The genuine novelty in crypto is not the linear (USD/USDT-margined) contract, which is plain generalized-BSM and falls straight out of the funding carry; it is the *inverse / coin-margined* contract, where the premium, the payoff and the P&L are all denominated in the base coin. An inverse option pays the usual option payoff divided by the terminal price, in coins per unit USD notional — and that $1/S_T$ factor makes the payoff a genuinely non-linear function of the terminal price with a real convexity/measure subtlety, not a rescaled vanilla. Celnet treats it as a distinct payoff shape with its own pricer derived from the share-/coin-measure numeraire change, gated three independent ways: a code-disjoint Monte-Carlo simulation (within reported standard error), an independent Gauss-Hermite quadrature of the raw payoff, and a structural sandwich ($V_{\text{inverse}} \cdot S_0 > V_{\text{linear}}$ for a convex payoff) that fails loudly on a naive rescale. The coin-measure delta/gamma/vega are reported in coins — a coin-margined desk hedges in coins.

Two contracts, one carry seam — and the inverse coin-settled subtlety priced precisely. The linear / USD(T)-margined contract is exactly generalized-BSM (the leaf assembles the funding carry $b = r - \text{funding}$, $F = S \cdot e^{(b \cdot t)}$, and calls the shared agnostic engine with zero new payoff math). The inverse / coin-margined contract is a genuinely new payoff shape whose $1/S_T$ factor — premium and payoff in the base coin — carries a real convexity/measure subtlety, priced by a numeraire change into the coin/share measure with coin-measure delta/gamma/vega rather than silently the USD ones. The 24×7 continuous trading clock (no weekend skip, no holiday centres, exchange-fixed UTC expiry) sits beside a three-way validation that always keeps an oracle able to disagree — an independent Gauss-Hermite quadrature, a code-disjoint Monte-Carlo of the raw inverse payoff within its reported standard error, and a structural sandwich bound — surfaced bit-identically across all five clients. The payoff math, measure correctness, convention identity and carry assembly are proven on the engine today; live venue connectivity, vol-surface/option-chain values, index-price fixings and funding-rate feeds are delivered through Celnet's implementation program on the same carry seam.

Settlement and convention depth, per class. Crypto runs a continuous trading-clock abstraction (every day a business day, no weekend skip or holiday centre) rather than special-casing crypto inside the FX calendar — FX keeps its business-day path byte-identical, and the crypto path simply never consults the weekend logic. Metals carry a lease rate as the carry leg modelled in place of a foreign rate, so the existing XAUUSD/XAGUSD path stays byte-identical while XPT/XPD and metal crosses become nameable, loco-London T+2 on the LBMA calendar. The crypto and equity smiles are strike/log-moneyness-axis (exchange option chains by strike), reusing the surface core's arbitrage-free interpolator and the SVI/SSVI/eSSVI parametric fits unchanged — the FX delta-axis machinery is simply not on those paths.

4.3 Linear products, structured additions & multi-dealer RFQ

The linear book. FX outright forward, FX swap (near + far legs) and the non-deliverable forward become *first-class priced products*, not merely derived forward math — a dedicated linear engine computes discounted-cashflow PV and the linear Greeks through the same carry seam. A forward's PV is $\text{side} \cdot \text{notional} \cdot \text{df} \cdot (F - K)$ with the forward rate produced by the carry — asset-class-agnostic, so an NDF on any future cross-asset underlying reuses the identical engine. The validation is deliberately non-circular (the discipline against a circular oracle that re-derives the same forward the production code uses): an external QuantLib forward engine as the independent route, plus structural identities a wrong implementation cannot satisfy — a forward struck at the fair forward has PV exactly zero, PV is linear in notional, a long-and-short forward at the same rate nets to zero, and an NDF and a deliverable forward of equal terms share the same risk-neutral PV when settled in the same numeraire.

LINEAR PRODUCT	PV	INDEPENDENT ORACLE
FX outright forward	$\text{side} \cdot \text{notional} \cdot \text{df} \cdot (F - K)$	QuantLib forward + fair-forward⇒PV=0 + notional-linearity + undiscounted short-tenor limit
FX swap (near + far)	Sum of two opposite-side outright forwards	QuantLib forward-points identity + near==far opposite-side netting to zero

LINEAR PRODUCT	PV	INDEPENDENT ORACLE
NDF	$\text{side} \cdot \text{notional} \cdot \text{df} \cdot (F - K)$ in the convertible currency	Hand-derived literal + NDF==deliverable-forward-PV in the same numeraire

Structured additions. Pivot — the one structured type the incumbent venue has that Celnet lacked — completes the structured catalogue: a TARF generalization with a pivot strike that switches the client between two enhanced rates, a cumulative knock-out target and adverse-leg leverage, modelled as a strip of periodic fixings on the existing Monte-Carlo stack (counter-based RNG, antithetic pairs, Welford standard-error accumulator, exact per-fixing GBM terminal law) with a reported price standard error. It is gated by a code-disjoint Monte-Carlo reimplementation *and* a degenerate limit: with the pivot set equal to the strike, the payoff is identically a TARF, asserted under common random numbers against the already-independently-gated TARF. Two further new payoff *shapes* ride the same carry seam — a perpetual option (Merton/McKean perpetual closed form, cross-checked by a long-but-finite American converging from below) and a listed-future option (Black-76, QuantLib-gated), with a cross-asset basket sandwich (worst-of $\leq$ single-leg $\leq$ best-of) proving the new shapes compose under the agnostic carry seam.

Multi-dealer RFQ aggregation. The single-dealer incumbents' true moat is the multi-bank venue, not product depth. Celnet's RFQ aggregation workflow fans one quote request to N quote sources concurrently — internal Celnet pricers *and* external liquidity-provider adapters over the existing FIX 4.4 engine — collects two-way bid/offer responses, ranks the best bid and best offer, applies a deterministic tie-break (earlier timestamp, then a stable provider id) and a timeout/last-look (non-responders dropped from the panel count; a winner past its validity window rejected and the next-best promoted), and returns an audited, ranked panel with consistent panel-count and won-by accounting. The aggregation, ranking, tie-break and last-look *algorithm* is built and validated against an injected ground-truth ladder over a ≥ 3 -provider loopback panel (so the oracle can disagree by construction). Live liquidity-provider connectivity over wide-area FIX and regulated-venue operation are delivered through Celnet's implementation program on the same finished seam — the multi-bank venue the single-dealer incumbents are built around, reached without a rewrite.

One ranked panel, every client. The SDK, CLI and GUI see the *identical* ranked multi-dealer panel — best bid/offer highlighted, the per-LP ladder, the won-by attribution — because there is exactly one aggregation engine behind one contract, with the internal pricer always present so the panel has at least one native dealer. The order path the venue would see is the order path the GUI sees.

The multi-bank-venue answer to a single-dealer incumbent — one ranked panel, every client. One client quote request fans out concurrently to N quote sources — an always-present internal Celnex pricer plus external FIX liquidity-provider adapters over the FIX 4.4 initiator engine — each returning a two-way bid/offer with a timestamp and validity window. The engine ranks best-bid (highest premium) and best-offer (lowest premium) with a deterministic earliest-then-stable-provider-id tie-break, drops timed-out non-responders and rejects last-look-stale winners (promoting the next-best), then books exactly the ranked winner over the same keyed line-bound token path as click-to-trade and FIX, with consistent panel-count and won-by accounting gated against an injected ground-truth ladder. The one ranked panel is bit-identical across GUI, SDK, CLI and Excel; the aggregation/ranking/tie-break/last-look algorithm plus FIX framing over a loopback initiator/acceptor pair are proven today, while live liquidity-provider connectivity and regulated-venue operation are delivered through Celnex's implementation program on the same seam.

4.4 Cross-asset risk: one firm book, every class

The risk layer generalizes the same way the pricing layer does: in place, FX byte-identical, additive everywhere else. The firm-wide position-fact cube already nets convention-canonicalized vanilla and exotic legs into a common numeraire; the cross-asset evolution widens the *fact* (an asset-class tag, a generic factor set, an underlying reference) while leaving the *algebra* untouched — because the additive-merge / non-additive-re-gather roll-up is already factor-generic. The non-negotiable invariant is that the existing FX firm-aggregate-equals-single-node parity stays byte-green to 1e-12 after the widening; the additive roll-up never reads the asset-class tag (it sums canonical Greeks regardless of class), and only the non-additive re-pricers route on the tag to the right asset-class engine — exactly as the cube already routes a barrier to the exotic pricer versus a vanilla to the FX engine.

The factor-keyed risk layer is where the "general meets zero-alloc" reconciliation pays off: the hot-core rate sensitivities are projected, off the hot path, into FRTB-bucketed factors — FX domestic/foreign rhos into the two legs' GIRR vertices; equity discount-rho and dividend-rho into GIRR plus equity-repo; commodity discount-rho and convenience into GIRR plus commodity carry. The FRTB-SA SbM machine — buckets, the three-scenario correlation max, within-bucket and cross-bucket roll-up and curvature — is already risk-class-agnostic (it takes caller-supplied risk weights and correlations as data, never compiled in), so the cross-asset extension adds the GIRR / equity / commodity / CSR bucket structure and a factor-projection, not a new kernel.

FRTB-SA RISK CLASS	BUCKET STRUCTURE	CROSS-ASSET ROLE
GIRR	Per-curve tenor-vertex delta buckets; intra-bucket ρ by tenor distance and same/different-curve, cross-bucket γ	The shared rate-risk vertices every class's discount/carry rho lands on — FX, metal, crypto, equity and commodity legs net here
Equity	The prescribed equity buckets (large/small cap $\times$ sector + indices), spot + repo sub-classes	Equity-option spot and repo factors
Commodity	The prescribed commodity buckets (energy/metals/agri/...), same-/different-commodity intra-bucket ρ	Commodity-option and metal carry factors

FRTB-SA RISK CLASS	BUCKET STRUCTURE	CROSS-ASSET ROLE
CSR non-securitised	Sector / credit-quality buckets	Delivered and parity-gated for cross-asset completeness; sourced when a credit-linked leg emits CSR factors

Every FRTB constant is re-derived from the BCBS primary text in the independent oracle (the discipline that caught a real circular-oracle bug in the FX low-correlation floor), the oracle is a longhand independent recomputation that never calls the production roll-up, and a constant-pinning gate fixes the correlation transform to hand-computed values with a negative control proving it fails on a wrong floor. The result is one cross-asset firm book — FX, metal, crypto, equity and commodity legs netting per factor into a single roll-up whose fan-out equals the single-node aggregate, with a real multi-class capital number rather than an FX-only approximation.

Every asset class nets into one firm book over factor-keyed facts. The pinned zero-allocation hot core emits the full FX desk Greek set in a fixed flat-data shape (delta, vega, discount-rho, carry-rho — a small fixed set, never an allocating factor map); a risk-layer projection *off the hot path* decomposes those rate sensitivities into FRTB-bucketed factors per class (FX, equity dividend/repo, commodity convenience-yield, crypto and rates on the same carry seam), with asset class named in exactly one place. Each leg flows as an immutable factor-keyed fact into the eight-axis hierarchical position-fact cube, where an additive per-factor roll-up that never reads the asset-class tag nets across all classes alongside non-additive VaR/ES and FRTB-SbM curvature re-gathered per node, projecting into the class-agnostic SbM machine over GIRR/FX/equity/commodity/CSR buckets, pruned by entitlements and a cascading limit tree, and collapsing to one cross-asset firm book exposed identically to GUI, SDK and Excel — proving firm-aggregate equals single-node to 1e-12 (1e-9 for re-gathered non-additive measures) with the existing FX parity staying byte-green. Dividend/repo/convenience/funding curves and FRTB risk-weights/correlations enter as caller-supplied data, sourced from the firm's curve estate; FX deliverable DRC is a documented, cited zero — there is no issuer jump-to-default in deliverable FX.

Carry-producing-market architecture · 6 asset classes (FX live + 5 cross-asset) · generalized-BSM with FX byte-identical · linear forward/swap/NDF · structured Pivot · multi-dealer RFQ · FRTB-SA GIRR/equity/commodity/CSR in one cube

The competitive consequence is that Celnet is not an FX-options platform with bolt-on asset classes, nor a generic multi-asset engine that loses the FX desk's conventions — it is a single coherent contract where FX is the zero-friction default, every other asset class is reached by the same carry seam with no payoff-engine rewrite, and every cross-asset claim is grounded in a validated method and an independent oracle. The architecture, the payoff math, the convention identity, the carry assembly and the aggregation algebra are proven and gated today; the non-FX leaves' live market connectivity — crypto and metal fixing values, funding-rate feeds, LP-panel sessions, dividend/repo and convenience curves — is delivered through Celnet's implementation program on those same finished seams. SynOption is a single-asset venue; Celnet is the whole asset spectrum on one engine.

5. State-of-the-Art Foundations & a Platform Built to Evolve

Why every method is the strongest published technique for its regime (the catalogue behind the \$3 pricing), the architectural SOTA leap of the carry abstraction, and the continuous-evolution model that lands new research as a composable leaf.

Celnet is state of the art on two axes at once: the **methods** it computes with and the **engineering** that lets those methods scale, compose and evolve without a rewrite. The numerics are not a fixed library frozen at a point in time — they are a catalogue of the strongest academically-grounded techniques for each problem, each cited to its method or paper, each validated against an independent oracle, and each reachable identically from every client. The engineering is not a monolith that ossifies — it is a layered, one-way-acyclic Rust workspace whose load-bearing seams (the carry-producing market abstraction, the one unversioned contract, the pinned zero-allocation hot core, the executable golden-vector conformance regime) are explicitly designed so the latest research lands as a **composable leaf or method** rather than a re-architecture. This section maps the SOTA methods catalogue, the engineering evolutions that make the platform world-class, and the continuous-evolution model that keeps it there.

The SOTA methods catalogue at a glance — a single in-core library spanning vanilla pricing and the full Greek set, the five-model smile and surface engine, the full exotic and structured catalogue and the LSV booking engine, each implemented with the strongest published method for its regime and continuously validated by independent-oracle parity checks and frozen golden tables. The cross-asset carry architecture and the Open Quant SDK are the seams through which new methods and asset classes compose in.

The academic-research-grounded methods catalogue

Every pricing, calibration and risk problem in Celnet is solved with the method the literature regards as state of the art for that regime, chosen for accuracy, robustness and scalability rather than expedience. Provenance lives in the doc comments and references — never in the public identifiers, which are purpose-named and vendor-neutral. The table maps each capability to its method and the published technique it descends from; the prose that follows draws out why each choice is the right one.

CAPABILITY	STATE-OF-THE-ART METHOD	REFERENCE / PROVENANCE
Vanilla FX pricing	Garman-Kohlhagen off the outright forward with dual discount factors	Garman-Kohlhagen; generalized-BSM $F = S \cdot e^{\{b \cdot t\}}$
Portfolio-scale Greeks	Reverse-mode adjoint algorithmic differentiation (AAD), $O(1)$ -in-factors	Adjoint AD
Smile / surface	Vanna-Volga, SABR, raw-SVI, SSVI, eSSVI (maturity-dependent ρ), arbitrage-free by construction	Gatheral-Jacquier SSVI; Hendriks-Martini eSSVI no-arb calendar

CAPABILITY	STATE-OF-THE-ART METHOD	REFERENCE / PROVENANCE
Local volatility	Dupire local-vol extraction feeding the exotic / PDE / LSV engines	Dupire
Local-stochastic vol	LSV with particle-method leverage calibration	Particle calibration of the leverage function
Stochastic vol (Heston)	Carr-Madan damped FFT integral $\approx$ Fang-Oosterlee COS, over a branch-cut-free characteristic function	Carr-Madan; Fang-Oosterlee COS; Cui-del Baño-Germano characteristic function
Arithmetic Asian	Turnbull-Wakeman moment-matching and Curran geometric-conditioning closed forms	Turnbull-Wakeman; Curran
Variance / volatility swaps	Log-contract $1/K^2$ static replication; Carr-Lee convexity adjustment for vol	Demeterfi-Kamal-Zou / Carr-Madan; Carr-Lee
American / Bermudan	PSOR free-boundary Crank-Nicolson FD and Longstaff-Schwartz LSM regression	Brennan-Schwartz PSOR; Longstaff-Schwartz
Correlated multi-asset	Cholesky-correlated GBM Monte-Carlo for basket / best-of / worst-of	Cholesky factorisation
Variance reduction	Sobol RQMC with Owen-style nested scramble and Brownian-bridge construction	Joe-Kuo direction numbers; Owen scramble; principal-bisection bridge
Regulatory capital	FRTB-SA Sensitivities-Based Method (SbM), curvature, RRAO, DRC	BCBS MAR21-23
Counterparty valuation	CVA / DVA / FVA over netting sets	XVA framework
Acceleration	Closed-form and path GPU kernels; pathwise / likelihood-ratio GPU Greeks	Pathwise & LR Greeks; cross-platform GPU (Metal / Vulkan / CUDA)

The choices are deliberate and defensible. The **volatility surface** uses five smile families because no single parameterisation serves a whole desk: Vanna-Volga reproduces the broker-quoted market, SABR and SVI/SSVI give parametric control, and **eSSVI** adds a maturity-dependent ρ with the Hendriks-Martini closed-form static no-arbitrage conditions — and because SSVI is byte-recovered as the constant- ρ special case of eSSVI, the strongest model is an additive extension, not a fork. **Heston** is priced two genuinely independent ways — a Carr-Madan damped Fourier integral and a Fang-Oosterlee COS expansion — over a **branch-cut-free characteristic function**, so the smile a desk calibrates is free of the integration artefacts a naive CF would induce. **American and Bermudan** options are solved by a PSOR free-boundary Crank-Nicolson finite-difference scheme and cross-checked by Longstaff-Schwartz least-squares Monte-Carlo regression, the two industry-standard early-exercise methods agreeing before a price is trusted. **Variance reduction** is not an afterthought: Sobol randomised QMC with an Owen-style nested scramble and a Brownian-bridge path construction delivers measured order-of-magnitude error reduction over plain Monte-Carlo on the exotic engines, with the direction numbers exposed for GPU reuse. And **Greeks at portfolio scale** are computed by genuine reverse-mode AAD — one reverse sweep yields the whole first-order block plus second-order gamma/vanna/volga — turning the cost of risk from $O(\text{factors})$ bump-and-

revalue into $O(1)$ -in-factors, the enabling technique for re-deriving a firm book's sensitivities in one sweep per position.

Method provenance, never in the API. The mathematical lineage of every method is recorded in code doc comments and the docs corpus — Garman-Kohlhagen, Gatheral, Hendriks-Martini, Carr-Madan, Fang-Oosterlee, Turnbull-Wakeman, Curran, Demeterfi-Kamal-Zou, Carr-Lee, Longstaff-Schwartz, Joe-Kuo, Owen, BCBS MAR. The public surface stays purpose-named and vendor-neutral (inputs are named for what they are, not for a person; a smile is selected as eSSVI, not a citation), so the strongest academic technique is delivered without leaking provenance into the contract.

The architectural SOTA leap: a carry-producing market abstraction

The single most consequential engineering evolution is the **cross-asset carry architecture**. It separates the one fused FX seam into three orthogonal layers — **identity** (an underlying-identity union naming asset class and settlement style), **market** (a carry that *produces* a forward and a discount factor rather than a variant payoff engines branch on), and **payoff** (asset-class-agnostic engines that consume the forward, the discount and a vol surface and nothing else). FX two-rate carry, equity dividend-and-repo, commodity convenience yield and crypto funding all collapse to the generalized-BSM forward $F = S \cdot e^{b \cdot t}$; they differ only in how the carry b is assembled from named factors. The binding **no-workaround test** makes this real: a payoff engine that branches on the carry variant is a review blocker, not a shortcut. This is the SOTA leap from an FX-options platform to a cross-asset derivatives platform — and it is built so the next asset class is a new carry-and-market builder with **zero payoff-engine edits**. The asset-class superset this unlocks, and its independent-oracle validation, are detailed in §4 (Multi-Asset).

The abstraction is reconciled with the hot core rather than fighting it. Because the Greek strip lives in the pinned zero-allocation hot core, it keeps a fixed flat-data shape — rate sensitivities are a small fixed set, not an allocating factor map — and the factor decomposition into FRTB GIRR / equity / commodity buckets happens off the hot path at the **risk layer**, whose roll-up algebra is already factor-generic. The FX relation is exact and recovered by the existing AAD adjoint. And the seam is in the right place by construction: a flat-rate carry is the degenerate single-point curve, so a future multi-curve OIS/CSA discounting model slots **behind the same forward/discount interface** with no payoff-engine change — the proof the abstraction generalises forward, not just sideways.

LAYER	SINGLE CONCERN	WHY IT IS SOTA
Identity	The only place asset class is named (FX, metal, digital-asset, equity, listed)	Additive growth; FX is the default, so existing tickets are unchanged
Market — carry	Produces the forward / discount from named factors	Two-rate / dividend-repo / convenience / funding all collapse to generalized-BSM
Payoff engines	Consume forward, discount and a vol surface only	A new asset class needs zero payoff-engine edits — the no-workaround test enforces it

LAYER	SINGLE CONCERN	WHY IT IS SOTA
Risk layer	Factor decomposition into FRTB buckets, off the hot path	Keeps the hot-core Greek strip fixed-shape and zero-alloc while staying factor-general

One unversioned contract, zero legacy — evolution in place

Celnet has exactly **one clean, current contract**. There is no schema versioning, no N/N-1 negotiation, no back-compatibility shim, and no mixed-version deployment window — a deliberate engineering stance that lets the platform evolve in place rather than accreting legacy. The vocabulary generalises by full cutover: the wire gains an underlying selector, a carry-model selector and carry-tagged rate sensitivities, with FX as the first-class default so every existing flow stays byte-equivalent after the legacy field is mapped into the new vocabulary. New payoff *shapes* (forward, swap, NDF, pivot, perpetual, listed-future option) get new product arms; existing arms become underlying-agnostic for free. Superseded code is deleted, not deprecated. The result is a contract that absorbs a new asset class or product as a purely additive change while remaining a single source of truth across all five clients — and a server-enforced **product × underlying validity matrix** that rejects an unsupported combination with a typed error rather than ever silently mis-pricing.

Executable conformance and independent-oracle verification

State-of-the-art numerics are worthless if they drift from their claims. Celnet's verification regime is itself engineered to SOTA: every capability claim is codified as an **executable check** against an **independent oracle that can disagree** — a QuantLib golden table, a closed-form limit, a published reference value, or a deliberately code-disjoint Monte-Carlo with its own RNG — so a correct-looking-but-wrong implementation surfaces loudly. The discipline is explicit about its own failure modes: the regime hunts **circular oracles** (a second implementation that re-derives the same algebra), re-derives regulatory constants from primary text, and requires at least one structural or limit gate per product that the production code cannot satisfy if it is wrong. The cross-client guarantee is mechanical: a language-neutral **golden-vector corpus** is generated by the engine, each value cross-checked by its oracle, and a conformance harness boots the **real edge** and asserts that every surface equals the frozen vector and is reachable from every client. A coverage lint ties each product on the wire to its vector and its independent check, so a new capability is not "done" until it is proven and reachable everywhere.

The anti-circular-oracle discipline. The hardest validation problem is an oracle that agrees with the code *because it makes the same mistake*. Celnet treats this as a first-class hazard: for every product the regime requires a route that can disagree — a code-disjoint Monte-Carlo for path-dependent payoffs, a QuantLib value for closed forms, a structural sandwich (worst-of $\leq$ single $\leq$ best-of) for baskets, a degenerate limit to an already-gated product. Regulatory constants are re-derived from the BCBS MAR text and pinned by a transform gate with a negative control, so a wrong correlation floor fails loudly rather than passing a circular self-check.

SOTA engineering: zero-alloc hot core, lock-free publication, consensus, sandbox

The runtime is engineered to the standard a mission-critical pricing platform demands. The **pinned zero-allocation hot core** does no allocation, locking or logging on the pricing path: wait-free single-producer/single-consumer rings feed pinned cores, and read-mostly state is published lock-free via an atomically swapped market-state snapshot and a single-writer seqlock top-of-book — telemetry offloads over a bounded, drop-on-full ring so observability is genuinely zero-cost on the hot path. Above the node, a **leader-replicated event log with full Raft** (election with Pre-Vote, conflicting-tail truncation, snapshot compaction and InstallSnapshot reseed) commits only on quorum and replays to bit-identical state. **Hot, blue-green zero-downtime state handoff** lets a new build take over a running book without dropping a price, and a **durable checksummed append-only journal** with compaction gives clean crash recovery. Extensibility is held to the same bar: the **deterministic plugin sandbox** runs Tier-2 WebAssembly models compute-budgeted with no ambient authority and supports bit-identical replay across runs, so user-extended analytics are reproducible and auditable. Determinism is end to end — the counter-based RNG is bit-identical between CPU and GPU, and the distributed substrate replays a workload containing a 1-ULP value so the bits must match exactly.

ENGINEERING PILLAR	MECHANISM	GUARANTEE EARNED
Zero-alloc hot core	Wait-free single-producer/single-consumer rings, pinned cores, atomically-swapped snapshot, single-writer sequence-lock	No allocation / lock / log on the pricing path; nanosecond-scale latency
Lock-free publication	Atomically swapped market state + sequence-lock top-of-book + bounded telemetry ring	Read-mostly state and zero-cost observability without contention
Consensus	Full Raft: election + Pre-Vote, truncation, compaction, snapshot reseed	Commit-on-quorum, bit-identical replicated replay
Hot upgrade / durability	Blue-green state handoff + checksummed append-only journal with compaction	Zero-downtime upgrades and clean crash recovery, bit-for-bit
Deterministic sandbox	Compute-budgeted, no-ambient-authority WebAssembly tier + bit-identical replay	Reproducible, auditable user-extended analytics

Built to absorb the latest research — continuous evolution

The deepest claim is not that Celnet is state of the art today, but that it is **built to stay state of the art**. The seams are deliberately placed so the strongest new technique lands as a composable unit rather than a re-architecture: a new asset class is a new carry-and-market builder and a new payoff leaf on the agnostic engines; a new smile model is an additive arm on the surface selector; a new exotic is a new product shape with its parity check; a new pricing model — even a bespoke proprietary one a client builds — is a sandboxed plugin behind the frozen extension contract. Multi-curve discounting slots behind the existing forward/discount interface. New regulatory calibrations are caller-supplied data, never compiled-in constants. Because the workspace is layered and one-way-acyclic with small disjoint components, this expansion fans out across parallel lanes without thrashing the frozen interface seams, and because every increment must pass the executable conformance and

independent-oracle regime before it is "done", growth never trades correctness for breadth. The cross-asset coverage already designed on this foundation — precious metals with a lease-rate carry, digital-assets including inverse coin-settled contracts with funding carry, equity dividend-and-repo, commodity Black-76, the linear forward/swap/NDF book, the structured Pivot, and multi-dealer RFQ aggregation — is the demonstration of the model: each is a leaf on the same architecture, validated by the same regime, reachable from the same five clients. The platform's design *is* its evolution strategy.

The evolution model in one view — one contract, the cross-asset leap, executable conformance, and the loop that keeps it state of the art. One clean, unversioned contract evolves in place with zero legacy (no schema versioning, no version negotiation, generalize-and-delete behind a byte-identical no-regression gate). The architectural leap to cross-asset rests on three orthogonal layers — an underlying-identity union as the only place asset class is named, a carry forward/discount producer where FX two-rate, equity (rate, dividend), commodity convenience and crypto funding all collapse to generalized cost-of-carry $F = S \cdot e^{(b \cdot t)}$, and asset-class-agnostic payoff engines, so a new asset class is a new carry builder with zero payoff-engine edits. An executable conformance regime of engine-generated golden vectors cross-checked by independent oracles (QuantLib golden tables, published closed forms, code-disjoint Monte-Carlo) feeds a five-client real-edge harness from which the parity matrix is generated. And a continuous evolution loop — critique lenses → compose the latest research as a method or disjoint leaf → verify against an oracle and parity gate → converge until two dry rounds — feeds back onto the same single contract without rewrite, every figure precisely labelled by its measurement context and every forward increment framed as a delivery step on a finished seam.

Celnet Capability Landscape

CELNET · CELER TECHNOLOGIES



The capability landscape — Celnet's quant, surface, risk, performance, distribution and extensibility capabilities form a composable superset on the carry-producing cross-asset architecture, with the one unversioned contract, the executable independent-oracle conformance regime and the pinned zero-alloc hot core as the seams through which the latest research evolves in as new leaves and methods rather than as a rewrite.

What makes Celnet state of the art. The methods are the strongest the literature offers for each regime, cited and oracle-validated. The carry-producing market abstraction makes the platform genuinely cross-asset with zero payoff-engine branching. The one unversioned contract evolves in place with zero legacy. The executable golden-vector conformance regime proves every claim, every change, against oracles that can disagree. The pinned zero-allocation hot core, lock-free publication, full Raft consensus and deterministic plugin sandbox earn performance, resilience and reproducibility. And the seams are placed so the next breakthrough is a composable leaf — so the platform does not merely meet the state of the art, it is engineered to keep pace with it.

6. Performance & Latency — Pricing Is Never the Bottleneck

The measured latency ladder, the architecture that keeps the core hot, the cross-platform GPU path — and (\$6.8 onward) the Open Quant SDK & tiered plugin host that runs a desk's private models inside the engine.

Celnet is engineered so that the maths is never what a counterparty waits on. The full FX desk valuation — price plus the complete 13-Greek set — runs in nanosecond-scale time on pinned, allocation-free hot cores; by the time a quote reaches the wire, network framing, not computation, is the only meaningful cost left. The decisive competitive point is not merely that Celnet is fast: it is that every latency figure below is *measured*, asserted by a regression gate that fails the build on breach, and labelled with exactly the scope it covers. Speed that is only asserted is no speed at all.

The headline. A complete price + 13-Greek valuation costs **42 ns at p50** and **125 ns at p99** on a single Apple M4 core — roughly **48× and 80× inside** the architecture's latency budget — sustaining **~13 million options/second/core**. These are not marketing numbers: a committed benchmark re-measures them on every run and fails the build if any percentile is breached.

6.1 The measured latency ladder

The headline budget is stated as **absolute per-option percentile ceilings**. The benchmark times *each individual* price-plus-full-13-Greek valuation — precisely the quantity the budget governs — into a coordinated-omission-aware HdrHistogram across 10 million timed samples after a 2 million-sample warmup, with priority elevation and (where the OS permits) core pinning.

In-core latency truth-gate — price + 13 Greeks, single Apple M4 core			
Metric	Budget	Measured (M4, single core)	Margin
p50	≤ 2 μs	42 ns	~48× inside
p99	≤ 10 μs	125 ns	~80× inside
p99.9	≤ 25 μs	~1.0–1.4 μs	~18–25× inside
throughput	≥ 1 M/s/core	~13.0 M opt/s/core	~13×

These are host-local, single-core figures — the intrinsic cost of the pricing core, not a wire latency. The platform reports its own measurement scope rather than flattering it: on Apple Silicon, macOS exposes no per-thread CPU affinity, so pinning is a no-op there and is reported as such while priority elevation still applies; on Linux CI both apply. The OS-jitter deep tail appears only at p99.99+, *deeper* than the gate, and is reported transparently rather than used to weaken it.

Fig 7 — The Celnet latency ladder. A pinned, allocation-free hot core (42 ns p50 / 125 ns p99 for price + 13 Greeks) feeds a wait-free single-producer/single-consumer ring that bridges to the async gRPC/WebSocket/FIX edge — leaving network framing, ~4 orders of magnitude above the in-core floor, as the only layer that materially matters.

6.2 Architecture that keeps the core hot

The nanosecond figures are a consequence of a deliberate two-tier design: a non-async, core-pinned, busy-poll pricing core, joined to a concurrency-rich async edge by wait-free single-producer/single-consumer rings. The critical path does nothing but compute.

LAYER	WHAT IT DOES FOR LATENCY
Pinned hot cores	Zero-allocation, lock-free, log-free valuation — the critical path only computes
Wait-free SPSC rings	Single-producer/single-consumer handoff bridges the async edge to the cores without contention
Read-mostly state publication	Atomically-swapped whole-market-state snapshots and a single-writer sequence-lock top-of-book let readers price without blocking writers
Lock-free broadcast fan-out ring	Each pair's tick is broadcast to all subscribers from one producer with exact, accounted conflation under back-pressure — never a torn read, never an allocation
Cache-padded counters	Telemetry and bookkeeping never cause false sharing against the hot path
Blue-green state handoff	Hot upgrades swap state with zero downtime, so performance is sustained across deploys

6.3 The surface-rebuild gate

Streaming a desk's smile in real time demands that the per-tick surface recompute also stays inside budget. A companion gate measures the sticky-delta recompute of all 11 standard tenors × 11 strikes for both the Vanna-Volga and SSVI models, asserting the **p99 ≤ 150 μs** budget. Measured: **~19.6 μs (VV) / ~7.8 μs (SSVI)** — about 7.7× / 19× inside. The one-off cold calibration (~2.2 ms VV / ~3.4 ms SSVI) is reported but deliberately *not* gated against the per-tick budget, because it runs on a quotes change, not on every spot tick — surfaced, never mis-gated against a budget that does not govern it.

6.4 A second, machine-independent gate: instruction counts

Wall-clock time floats with host load, so a CI gate on it must carry a generous tolerance. Celnet adds a complementary **instruction-count gate** that measures a **deterministic, machine-independent** quantity — retired instructions, cache accesses and estimated cycles per pricing call — over price, the price-plus-Greeks valuation, and a 64-strike batch slice. That makes it the right primitive for catching a *code-level* regression — an extra branch, a lost inlining, an accidental allocation — the moment it lands, with a tight baseline and no flake.

The instruction profiler is Linux-only, so this runs as a dedicated Linux CI lane and never affects local development on the M4 dev host.

6.5 The cross-platform GPU path — correctness and ratios

For batch revaluation, Celnet drives a cross-platform GPU backend (Metal / Vulkan / DX12) whose results are reconciled **node-by-node** against the f64 CPU oracle and the QuantLib golden tables, using a bit-identical counter-based RNG implemented byte-for-byte on both CPU and GPU. The GPU load harness sweeps the batch size and reports per-dispatch latency tails plus a **host-local GPU/CPU throughput ratio**:

M4 Metal dispatch-amortization curve — host-local GPU/CPU throughput ratio (sweep 4 096 ... 1 048 576 paths)		
BATCH (PATHS)	GPU/CPU RATIO	REGIME
4 096	0.68×	Dispatch overhead dominates small batches
262 144	~16.5×	Committed CI-sized baseline, regression-gated
1 048 576	~53×	Device saturated — the GPU pulls decisively ahead

When no GPU adapter is present the backend falls back transparently to the CPU oracle (a software rasteriser in headless CI), and the run is labelled a completes-within-ceiling check rather than a speedup claim. The GPU estate is broad: a closed-form batch kernel, a multi-step path-dependent kernel, pathwise/LR Greeks, and an Owen-scrambled Joe-Kuo Sobol' QMC path (~37.7×/88.5× measured variance reduction) reused across exotics, XVA and the GPU kernels — all reconciled three-way GPU-f32 ≈ CPU-f64 ≈ golden within a *derived* (not asserted) f32 bound, with a CPU↔GPU Sobol' known-answer test.

6.6 The wire-path and observability

The in-core figures are the *compute floor*; the figure a counterparty actually observes is the wire-path round-trip. The wire harness spins the real server in-process on a loopback port, opens concurrent RFS streaming subscriptions as sustained background load, and times two-way request round-trips through the full gRPC/HTTP-2 stack, codec, the wait-free edge-to-core hop, and pricing. The wire-path sits ~4 orders of magnitude above the in-core floor — dominated by framing, not by the model — which is exactly the design intent.

Mission-critical visibility costs nothing on the hot path. Latency is captured into coordinated-omission-aware tail-percentile histograms, and those same percentiles are surfaced live to clients on the stream heartbeat (server-side p50/p99/p99.9 in nanoseconds, plus conflation-drop counts). Telemetry leaves the hot core over a **bounded, drop-on-full ring**: under pressure the platform sheds observability data, never throughput. The pinned core stays log-, lock-, and allocation-free; everything instrumental is offloaded. All of this is held by a benchmark gate that runs four arms on every change — the in-core absolute latency gate, the surface-rebuild absolute gate, the wire-path relative-regression gate (fail on >2× baseline), and the fleet loopback SLO gate — all of which must pass.

6.7 The performance delivery program on the engine

The gates establish the in-core latency truth-gate, the surface-rebuild gate, the instruction-count gate, and the loopback wire/fleet benches as continuously-proven properties — the measured compute foundation. Three classes

of absolute headline are delivered through Celnet's implementation program, each lit up against the target hardware and network topology on a finished seam:

- **Cross-host wire p99 and the SOTA kernel-bypass NIC datapath** — delivered on production cross-host topology; the loopback benches already establish the compute + framing lower bound the program builds on.
- **The absolute wire-latency SLOs** — delivered against the production network fabric.
- **CUDA/NVIDIA absolute GPU throughput, the ≤ 50 ms exotic target, and the workload absolute figures** — delivered through the CUDA backend on NVIDIA hardware. GPU correctness and host-local ratios are proven today (Apple M4 Metal lacks f64), with the f64-correct kernels already in place so the program lights up the absolute throughput headline on the target accelerator.

Network framing — not computation — is the only meaningful latency floor, and Celnet states it as a *shape* backed by measured ratios and a SOTA kernel-bypass datapath, never as a fabricated cross-host number.

6 (cont.) · Extensibility — The Open Quant SDK & Tiered Plugin Host

The user-extensibility edge the closed venues structurally lack — frozen model traits, a self-describing registry, and a tiered sandbox so a desk extends the already-deep catalogue with private IP, without forking Celnet.

A pricing platform is only as good as the models a desk can put inside it. Celnet ships a functionally complete analytics estate already — vanilla, the full first-generation exotics, structured and path-dependent products, American/Bermudan early exercise, correlated multi-asset baskets, an LSV booking model and a standalone Heston engine, all on one wire — yet treats the desk's *own* intellectual property as a first-class extension of that estate. Proprietary models run **inside the engine**, on the same registry and the same routing as the built-in analytics, without ever leaving the firm and without forking the platform. The vehicle is the **Open Quant SDK** — one frozen, runtime-agnostic contract — served by a **tiered plugin host** that trades performance against isolation while keeping every shipped tier deterministic and bit-identical on replay.

The structural edge. The deep-catalogue incumbents make a desk choose: a closed terminal gives you the vendor's models; a modern library gives you code but not an engine. Celnet gives both — the desk's own models, at native speed where they are trusted and under hard deterministic isolation where they are not, all through one clean frozen contract, layered over a catalogue that already matches what the front-to-back platforms charge for.

Fig 5 — The tiered plugin host. The Open Quant SDK is a single frozen contract; a tier-blind registry routes a registered model to the tier its trust level warrants. The native (Tier-0) and WebAssembly-sandbox (Tier-2) tiers are shipped and gated today; the signed-shared-object (Tier-1) and OS-sandbox (Tier-3) tiers are seamed behind the identical contract and delivered through Celnet's implementation program — a user-extensibility surface the closed single-dealer venues cannot match.

6.8 The Open Quant SDK — one contract, any model

The SDK is a small, stable set of contracts describing the three things a model can be, plus the metadata and discovery that let the engine route to it without hard-wiring:

SDK SURFACE	WHAT A QUANT IMPLEMENTS	WHAT CELNET DOES WITH IT
Pricing model	Price and the full Greek vector for an instrument	Routes pricing and risk through the model exactly as it does the native vanilla and exotics engines
Smile model	A smile/surface parameterisation (extends the core smile seam)	Marks, recalibrates, and arbitrage-gates the model alongside the built-in Vanna-Volga, SABR, SVI, SSVI and eSSVI families

SDK SURFACE	WHAT A QUANT IMPLEMENTS	WHAT CELNET DOES WITH IT
Calibration	A fit from market observables to model parameters	Drives the model from the same broker risk-reversal/butterfly/ATM and market-series inputs the built-in calibrators consume
Model descriptor / registry	A self-describing manifest (model id, kind, Greek support)	Lets the engine discover, name, and route the model by purpose with no hard-coded wiring
Guest-interface mirror	—	Mirrors the contract one-to-one for the sandboxed tier so a guest model speaks the identical shape

Because the contract is **frozen and unversioned**, a model written against it today keeps working as Celnet evolves — there is exactly one current contract, never an N/N-1 negotiation. A registered model is indistinguishable, at the call site, from a built-in one: the host registry is **tier-blind** — native, WebAssembly and loaded models all land behind one model lookup — so the engine routes a private smile model or a bespoke pricer through the same path that serves the standard catalogue. The desk uses the SDK to *extend* the shipped core with bespoke and structured products the firm wants to keep private — not to reach the catalogue, which is shipped.

6.9 The tiered host — performance and isolation, never one or the other

Not every model carries the same trust. A house model authored by the firm's own quants wants native speed; a model under evaluation, or one carrying third-party code, wants containment. The host is a ladder of tiers behind the **same** contract, so the choice is a routing decision, not a rewrite.

TIER	EXECUTION	ISOLATION POSTURE	TYPICAL USE	STATUS
Native (Tier-0)	In-process, native speed, monomorphized zero-alloc dispatch	Trusted, first-party	House models on the hot path	Shipped
WebAssembly (Tier-2)	Compute-budgeted WebAssembly interpreter	No ambient OS access; capability-scoped	Untrusted or under-evaluation models	Shipped
Signed shared-object (Tier-1)	Loaded native object via a stable ABI	Cryptographically signed, publisher-allowlisted	Vetted partner models needing native speed	Implementation program
OS-sandbox (Tier-3)	Native under OS confinement	Kernel-enforced (Landlock / seccomp), Linux-only	Native models needing belt-and-braces containment	Implementation program

The host runs the native and WebAssembly tiers, both gated; the signed-shared-object and OS-sandbox tiers are specified and seamed behind the identical frozen contract and delivered through Celnet's implementation program, slotting in without a contract change.

Tier-0 — native adapts any first-party pricing model to the tier-agnostic host seam. It is generic over the concrete model, so the call is a direct, monomorphized, inlinable dispatch with **zero per-call allocation** — the hot-path property Tier-0 exists to preserve. There is no sandbox here by design; Tier-0 is trusted first-party code.

Tier-2 — the WebAssembly sandbox is the centre of gravity for untrusted code, and the advisory-clean choice of a pure-Rust interpreter (the originally-planned runtime was blocked under the open-source-only policy by 2026 security advisories). A user model compiled to a core WebAssembly module runs with **no ambient operating-system authority** — no system interface, no clock, no RNG, no threads, no filesystem, no network. The capability boundary is deliberately narrow: the *only* imports a guest may resolve are five host-provided mathematical primitives — `exp`, `ln`, `sqrt`, the normal density and the normal CDF — each routed through the same platform-independent maths library the native tier uses. Any import outside that set is **denied at link time** with a precise, attributable error before the module is even instantiated.

The sandbox is hardened beyond capability-denial alone:

- **Compute SLA.** Every guest call runs under a per-call **compute budget** (a deterministic instruction-cost counter, not wall-clock). Exhaustion interrupts the interpreter at the budget and yields a typed error — never a hang or panic. Even a module's initialiser is budget-bounded at load, so a runaway start cannot hang the host.
- **Resource ceilings.** A store-level limiter caps linear memory (64 MiB), table elements, memories, tables and instances; an over-budget memory or table growth traps deterministically rather than returning a sentinel.
- **Narrowed feature set.** The interpreter disables WebAssembly proposals a pricer never needs, so a guest using them fails validation rather than widening the attack surface.
- **Boundary determinism.** Every floating-point value crossing the boundary is NaN-canonicalised, so a malformed number can never leak non-determinism into the engine.
- **No unsafe code.** The whole plugin host forbids unsafe code; the interpreter is the only execution substrate, driven entirely through its safe API.

Four work-stream gates prove these properties: **capability-denial** (an unknown import fails to load), **compute-budget bounded** (a runaway guest traps within budget and does not hang, verified including the initialiser path), **replay bit-identity**, and **Tier-0 == Tier-2 interchangeability** (a native model and its WebAssembly twin agree to the last bit through one tier-blind registry). A guest's own domain rejection surfaces as a model error, not a sandbox fault — so a Tier-2 verdict is indistinguishable from a Tier-0 one.

6.10 Determinism, replay, and the arbitrage self-check

Determinism is a platform guarantee, not a hope. A single Tier-2 model produces **bit-identical** results on replay — the same inputs reproduce the same outputs to the last bit (including any canonicalised NaN), run after run, on every platform the interpreter runs on — which is what makes plugin-priced trades auditable and reproducible across the firm. The host carries a deterministic replay harness that replays a fixed market snapshot through a model and asserts bit-identity across runs; the same compute budget interrupts at the same instruction, so even a near-budget model replays identically. Cross-platform identity additionally requires the guest's transcendentals to be the platform-independent library ones — which is precisely why the host exposes them as capabilities rather than letting the guest reach for a hardware intrinsic.

Bit-identity *between* a native and a WebAssembly implementation of the same pricer holds only when both perform the same sequence of IEEE-754 operations through the same maths library. The harness proves the strong bit-exact claim for a deliberately op-order-matched twin; for an arbitrary pair, agreement is within the model's documented numerical tolerance, and a tolerance-based comparator is the right gate. Celnet states this distinction with engineering precision — the exact guarantee a firm's audit and reproducibility regime depends on.

A smile model loaded through the SDK is held to the same standard as the built-ins: a **static (butterfly) no-arbitrage self-check** is built into the smile-model contract — a model-free density-positivity test over the strike grid, in the spirit of the Carr-Madan call-spread conditions. A plugin therefore cannot quietly publish an arbitrageable surface; an arbitrageable grid is rejected with a typed error.

6.11 What it means for the desk

- **Private IP stays private.** Proprietary pricers and smiles run *inside* Celnet's engine — sandboxed, deterministic, hot-loadable — and the source never leaves the firm.
- **Model evolution without downtime.** New or revised models load into the running engine and inherit Celnet's zero-downtime blue-green state handoff; there is no rebuild-and-redeploy of the platform to ship a model change.
- **No fork, ever.** The frozen SDK means a desk extends Celnet rather than maintaining a private branch — bespoke coverage is additive on top of an already-complete built-in catalogue, and upstream improvements arrive cleanly.
- **One discipline for every model.** House and experimental models live behind the same contract, the same arbitrage gates, the same replay guarantee — so a model's trust tier changes how it is *contained*, never how it is *called* or trusted to be correct.

7. The API Contract, Cross-Client Parity, Excel & the Trader GUI

One unversioned contract consumed identically by five clients — the five service families, the streaming session, the SDK/CLI, the Excel add-in and the trader GUI — so a value is bit-identical wherever it is read.

Most derivatives desks run on a patchwork: a pricing terminal, a separate risk system, a vol-marking tool, a spreadsheet bridge, and a bespoke GUI — each with its own version negotiation, its own numbers, and its own opportunities to disagree. Celnet collapses that estate onto **one unversioned wire contract** consumed identically by five clients. There is no schema-version negotiation, no N/N-1 compatibility window, no privileged internal path — a single current contract that every surface speaks. This is the governing **API-first** rule: capability lives in the contract, never in a client, so the number a trader reads in the GUI is bit-identical to the one the Rust SDK returns, the CLI prints, and a spreadsheet cell spills.

What runs over that contract is the *full* FX-options catalogue — vanilla, the complete first-generation exotics, structured and path-dependent products, American/Bermudan early exercise, correlated multi-asset baskets, and an LSV booking model — reachable from all five surfaces, with bit-identical values proven by an executable client-parity matrix. The front-end is just another client.

One contract, every surface. The trader GUI, the typed Rust SDK, the admin CLI, and the Excel **CELNET.*** add-in are peers over the same five service families — Pricing, Quote, Stream, Risk, Surface. A value is identical wherever it is read, across both the binary gRPC edge and the byte-identical WebSocket JSON mirror.

7.1 The Five Service Families

The five gRPC service families that map to how a desk works — price, RFQ lifecycle, stream, firm-scale risk, surface marking — plus the unified instrument that carries every product on one wire.

The contract is organised into **five gRPC service families** that map directly to how a desk works: get a price, run an RFQ lifecycle, stream live markets, query firm-scale risk, and mark surfaces. Every operation is enumerated below.

SERVICE FAMILY	OPERATION	PURPOSE
Pricing	Price	One-shot price plus the full 13-Greek risk set for any instrument, against live or a pinned marked surface; the response carries a correlation id, the surface version it priced on, and a price standard error (set only for MC products).
Quote	Request quote	Request a firm two-way (RFQ): bid/mid/offer with a quote id, a last-look validity window, Greeks, attribution, and an MC std-error when applicable. Idempotent on the request key.
	Accept quote	Accept a live quote and book the resulting execution.
	Reject quote	Decline to trade; returns a typed acknowledgement, never an execution.
Stream	Stream session	One bidirectional, multiplexed channel carrying many concurrent subscriptions — streaming prices, two-ways, observable series — with sequence-gap resync, in-place modify, and click-to-trade. A blotter watching hundreds of structures uses one session.
Risk	List positions	List the open positions the cube aggregates (scoped + entitlement-pruned), each carrying its attribution.
	Aggregate risk	Roll the entitled facts up over an org dimension into a node tree of netted additive + re-derived non-additive measures in a reporting numeraire.
	Drill risk	Drill one node into its child sub-nodes and/or contributing positions (the Book → Risk drill), entitlement-pruned.
	Limit status	Read the limit tree + utilization / RAG status for a scope node, including the hard-breach escalation flag.
Surface	Get smile	Read the calibrated smile for one (pair, tenor) on the delta axis.
	Mark surface	Mark / recalibrate the surface for a pair from a broker quote set, under a chosen smile model, depositing it under a fresh surface version.
	Scenario	Reprice an instrument across a spot / vol / rate shock grid.

Aggregation, common-numeraire conversion, non-additive re-derivation, entitlement pruning, and limit RAG are all the server's job — a client never loops positions and sums.

The unified instrument — one vocabulary, every product

A **single instrument type** runs through Pricing, Quote, Stream, and Scenario — the same vocabulary describes a vanilla, a digital, a barrier, a TARF, a basket, or an American option, so the contract does not fork by product. The payoff is a single selectable product with **19 arms**; each Monte-Carlo-priced arm carries a price standard error and is never presented as machine-precision.

PRODUCT	ENGINE / METHOD	MC STD-ERROR?
Vanilla European	Garman-Kohlhagen closed form (analytic, or LSV via the pricing-model directive)	—
Multi-leg strategy (RR / strangle / straddle / seagull)	Sum of vanilla legs (analytic)	—
Single-barrier knock-in / knock-out	Analytic + CN-Rannacher PDE / Philox MC, VV survival overlay (or LSV)	— / MC
Double-barrier	Analytic / PDE / MC	— / MC
Digital (cash-/asset-or-nothing)	Closed form	—
One- / no- / double-no- / double-one-touch	Closed form + VV overlay	—
Variance swap	Log-contract static replication (Demeterfi-DKZ / Carr-Madan)	—
Volatility swap	Carr-Lee convexity-adjusted fair-vol strike	—
Fixed-strike arithmetic-average Asian	Turnbull-Wakeman / Curran (closed form)	—
Forward-start vanilla (strike reset)	Rubinstein dual-carry closed form	—
Cliquet / ratchet (plain or clamped)	Σ forward-start legs (closed form), or clamped MC	clamped
Quanto (vanilla or digital)	Settlement-currency-converted closed form	—
Target-Redemption Forward	Geared-fixing knock-out Monte-Carlo	yes
Accumulator (pivot accrual + up-and-out)	Monte-Carlo	yes
Lookback (floating / fixed strike)	Continuous closed form, or discrete MC	discrete
Window knock-out (active inside a calendar window)	LSV ADI-PDE (or MC); no closed form	MC
American / Bermudan early-exercise vanilla	PSOR free-boundary FD (default), or Longstaff-Schwartz LSM	LSM

PRODUCT	ENGINE / METHOD	MC STD-ERROR?
Correlated multi-asset (weighted basket / best-of / worst-of), N legs	Cholesky-correlated multi-asset GBM MC over scrambled-Sobol / Brownian-bridge	yes

The catalogue is full: it spans vanilla → the complete first-generation exotics → structured / path-dependent → American/Bermudan → correlated multi-asset basket → a particle-calibrated LSV booking model.

Two first-class directives on every instrument

- **The booking-model selector** — the default routes to the product's native analytic / closed-form engine and is byte-identical to the contract before the directive existed. The local-stochastic-vol setting routes the supported products (vanilla, continuously-monitored single-barrier knock-out, window-barrier) through the LSV booking engine — a particle-calibrated leverage surface over a stochastic-variance backbone on a 2-D ADI PDE. Selecting LSV for any other product is a hard invalid-argument error, never a silent fallback. The directive travels on the instrument, so it reaches price / quote / stream / scenario uniformly.
- **The smile-model selector** carried on the mark-surface and scenario requests: **five** calibration families — market-hedge (vanna-volga, the default), stochastic-vol (SABR), parametric (SVI), parametric-surface (SSVI), and extended-surface (eSSVI — maturity-dependent correlation, closed-form arbitrage-free). The contract also carries an **attribution identity** (book / owner / attribution record) keying every price and position to its place in the risk hierarchy.

Numerical disclosure as a contract feature. The price standard error on the price and quote responses is presence-tracked: set only for Monte-Carlo-priced products (TARF, accumulator, discrete lookback, clamped cliquet, basket, LSM American), absent for closed-form products whose price is exact. The same disclosure crosses gRPC, the WebSocket mirror, the SDK, the CLI, and Excel — a client never mistakes an MC estimate for closed-form precision. The XVA engine computes CVA/DVA/FVA over netting sets on the risk-neutral GBM exposure pipeline; Celnet's implementation program extends it onto live CSAs / collateral / wrong-way risk on the same pipeline.

Surface-version pinning

Pricing and risk are only reproducible if everyone agrees on the surface. Every marked surface is deposited under a fresh surface version, and any pricing, RFQ, or stream request can **pin** to a specific version. An unknown version is rejected — never silently resolved to the live surface — so a quote, a risk report, and a re-priced ticket all reference the identical calibrated smile. The surface version is a *data* field, echoed on the price response, the quote, and the heartbeat, never an API version.

7.2 The Multiplex Streaming Session, Market-Series Feed & Click-to-Trade

One bidirectional session carrying many subscriptions with per-subscription sequencing and resync, keyed-MAC click-to-trade with last-look, and the byte-identical WebSocket mirror of the whole contract.

The streaming session is a single bidirectional channel that fans out to many subscriptions. The client sends control messages (subscribe / modify / unsubscribe / resync / execute / heartbeat / market-series subscribe); the server returns snapshot / update / heartbeat / stream-end / executed / stream-reject / market-series frames. Each price subscription follows a clean lifecycle — **Subscribe** → **Snapshot** → **Update(seq)** → **Heartbeat** — with monotonic sequence numbers so a client can detect a gap and issue a resync to re-baseline, and an in-place modify that re-bases a subscription (new strike, notional, or tenor) with a fresh snapshot without tearing it down.

Market-series feed — multiplexed on the same session. A market-series subscribe opens a series subscription in the same subscription-id space, streaming one of five labelled, unit-bearing observables: ATM vol, spot, risk-reversal, butterfly, forward. This is the contract the GUI trend modes consume — no abstract index, every point carries its unit.

Zero-cost observability on the liveness beat. The heartbeat carries server-side telemetry read off the *drain side* — the pinned zero-alloc pricing core stays untouched: a conflation-drop count (the exact skip count from the broadcast ring's received + skipped == produced accounting), a histogram of server-side compute latency (p50 / p99 / p99.9 in nanoseconds), the pinned surface version, and the sequence number. A consumer can watch its own tail latency and conflation without a side channel.

Typed terminal reasons. A subscription that ends sends a stream-end with a typed reason (`LAGGED` / `DRAINING` / `UNSUBSCRIBED` / `EXPIRED`), and a declined click sends a stream-reject with a typed reason (`EXPIRED` / `UNKNOWN_TOKEN` / `ALREADY_CONSUMED`) — never an opaque string.

Click-to-trade is built into the same channel. A streamed line carries an unguessable, keyed-MAC tradable token — one to sell at the bid, one to buy at the offer — cryptographically bound to that exact line. To deal, the client returns the token through an execute message:

- **Last-look validity** — the token carries a validity window; a stale token is declined (expired) rather than filled at a moved market.
- **Forgery-proof** — the token is a keyed message authentication code over the line-binding tuple under a per-session secret; a forged or tampered token is rejected (unknown token).
- **Idempotent** — a token is single-use; a duplicate or already-consumed execute is rejected (already consumed), so a retransmit can never double-deal.

The multiplex streaming session. Many price + market-series subscriptions over one channel, sequence-gap Resync, in-place Modify, Heartbeat observability (conflation drops + p50/p99/p99.9), and the keyed-MAC click-to-trade token with last-look and idempotent Execute. One session powers a blotter watching hundreds of lines.

The byte-identical WebSocket mirror

The contract is served over a high-performance binary gRPC transport for native clients **and** over a byte-identical WebSocket JSON mirror for browser and lightweight clients. The mirror covers all five service families, field-for-field — it is not a parallel API but the same contract, so a value crossing the WebSocket is identical to the one crossing the binary edge. This is what lets the React GUI and a native SDK client share one mental model and one set of guarantees, and the JSON projection is continuously checked to mirror the wire contract exactly. (The in-core latency truth-gate plus loopback benches establish the compute foundation; the absolute cross-host wire p99 and the SOTA kernel-bypass NIC datapath are delivered through Celnet's implementation program on production topology.)

7.3 The Typed SDK, Instrument Builder & Admin CLI

A typed Rust client with reconnect-liveness and an instrument builder for every product, plus an admin CLI — the same contract a desk scripts against and the GUI consumes.

The **typed Rust SDK** turns the contract into ergonomic, statically-checked calls grouped by service family (about twenty public methods):

- **Pricing & Quote** — price; request a quote (with optional attribution), accept it, reject it.
- **Surface** — read a smile, mark or recalibrate a surface (optionally under a chosen model), and run a scenario (optionally with a chosen model and with risk decomposition).
- **Risk** — list positions, aggregate risk, drill a node, read limit status.
- **Stream** — open a session, then subscribe to prices, attributed prices or observable series; per subscription, read the next event, execute a click-to-trade, and unsubscribe.

It returns **typed errors** (a rejected stale token, an unknown surface version, or a typed expired / unknown-token / already-consumed outcome surface as specific, matchable variants — never strings) and has built-in **reconnect-liveness**: across a blue-green cutover or a dropped link the SDK auto-resyncs subscriptions on the same session and drains every pending click-to-trade waiter with a typed outcome, so a client never hangs on a lost connection.

The **instrument builder** constructs every product arm with a fluent, typed constructor — vanilla, multi-leg strategy, single and double barrier, digital, the four touch variants, variance and volatility swaps, Asian, forward-start, cliquet, quanto, TARF, accumulator, lookback, window-barrier, American/Bermudan and correlated basket — plus the booking-model directive and a local-stochastic-vol convenience. Three runnable examples ship: pricing an exotic, quote-and-trade, and a streaming blotter.

The **admin CLI** drives the same contract through the SDK — no separate control API to learn. Seven top-level subcommands:

COMMAND	DRIVES	NOTES
price	Pricing	Vanilla GK price + full Greek set.
surface	Surface	Build a smile from broker quotes; print vols + the arbitrage report.
exotic	Pricing	The exotic catalogue — 16 sub-variants : vanilla, digital, one-touch, double-no-touch, barrier, window-barrier, var-swap, vol-swap, Asian, forward-start, cliquet, quanto, TARF, accumulator, lookback, American. A model flag routes the supported products to the LSV engine.
basket	Pricing	Correlated multi-asset (weighted basket / best-of / worst-of) over N legs; reports a std-error.
convention	resolver	Resolve and print the convention record for a pair + tenor.
risk	Risk	Four sub-subcommands aggregate / drill / positions / limits, with entitlement grant / deny scope flags — the same risk service the GUI Book view and Excel consume.

COMMAND	DRIVES	NOTES
<code>stream</code>	Stream	Subscribe to a two-way RFS stream, print sequenced ticks, unsubscribe cleanly.

Because there is exactly one contract and every client is a peer over it, parity is **structural rather than aspirational**, proven by an executable client-parity matrix: all 18 product families × the five service families are reachable from all five surfaces, with stated, documented notes — correlated-basket Greeks are zeroed beside the real measured premium until the basket-Jacobian increment of Celnet's implementation program lands on the pathwise/AAD machinery, and the LSV booking model is reached *through* a priced product, not as a raw calibration call. Each "n/a" states its reason; no cell is a stub.

7.4 Excel: 27 CELNET.* Functions, One Pricing Path

A first-class spreadsheet client where no pricing happens in the cell — every number is the server's value, with the same conventions and std-error discipline the GUI and SDK receive.

The spreadsheet is where structuring, marking, and ad-hoc risk actually happen on most desks — so Celnet meets the desk there, natively. An Office.js add-in exposes the platform's full pricing, surface, quoting, streaming, and server-side risk capability as worksheet functions under the **CELNET.*** namespace, speaking the *same* single, unversioned contract as the GUI, the SDK, and the CLI. Every number a cell shows is the server's value, computed in the engine and bit-identical to the GUI. There is no second pricing path hiding in a macro.

This is not a thin "price a vanilla in a cell" plug-in. The add-in ships **27 CELNET.*** worksheet functions covering every product on the wire — vanilla, the full first-generation exotics, structured and path-dependent products, American/Bermudan early exercise, and correlated multi-asset basket / best-of / worst-of — plus surface marking under five smile families, RFQ, live streaming, the firm's server-side hierarchical risk roll-up, and live server observability, rendered into native spilled dynamic arrays.

The branded Excel surface. Live two-way prices, spilled Greeks, a smile grid, and streaming observables — all served from the engine over the one contract. The cell is a view onto the firm's system-of-record, never a parallel calculator.

Three rules that make the add-in safe on a live desk

- **No pricing in the cell.** The add-in never approximates, never falls back to a local model, never reimplements a Greek. Each function marshals inputs to the engine and renders what comes back — there is no numerical code in the add-in.
- **Read is zero-click; write is two-phase.** Read-and-spill functions evaluate the moment the cell does. Anything that mutates shared state — contributing a mark — stages the intent (rendering **PENDING**) and transmits only after explicit task-pane confirmation. Recalculating a sheet can never accidentally publish a price.
- **Convention and version transparency, per cell.** Every priced cell carries a spilled footer with the exact convention basis (delta convention, ATM rule, premium style, day-count, settlement) and the surface version it was marked against, plus a capture timestamp (`conv: Δ ... | ATM ... | ... | surface v... | t=...`).

Failures are typed, not silent: a bad-argument error (`#CELNET_ARG!`) is distinct from a generic transport error (`#CELNET_ERR!`), and streaming cells flip to a visible (`stale`) / (`resync`) state on a heartbeat gap rather than freezing as live.

The complete `CELNET.*` reference

Pricing & vanilla Greeks

FUNCTION	SIGNATURE	BEHAVIOUR / SPILL
<code>CELNET.PRI</code> <code>CE</code>	<code>(pair, tenor, strikeOrDelta, callPut, notional)</code>	Prices a vanilla off the engine; returns the mid of the two-way. Strike may be absolute (<code>1.12</code>) or a delta (<code>25dP</code> , <code>ATM</code>). Single mid premium.
<code>CELNET.GRE</code> <code>EKS</code>	<code>(pair, tenor, strikeOrDelta, callPut, notional)</code>	The full 13-Greek vector in one pass. 14×2: 13 <code>[name, value]</code> rows + convention footer.
<code>CELNET.RF</code> <code>Q</code>	<code>(pair, tenor, strikeOrDelta, callPut, notional)</code>	Requests a two-way; the <code>quoteId</code> binds the task-pane Trade ticket and carries a last-look validity. 1×N: <code>[bid, offer, quoteId, validUntil]</code> + footer.

Exotics & structured products

Each exotic spill is a premium/PV row, an optional standard-error row (only when MC-priced), the 13 risk Greeks, and the convention footer. Every premium is the server's value, bit-identical to the SDK/CLI.

FUNCTION	SIGNATURE	ENGINE / STD-ERROR
<code>CELNET.BAR</code> <code>RIER</code>	<code>(pair, tenor, strikeOrDelta, callPut, notional, barrier, [kind], [side], [upperBarrier], [rebate], [monitoring], [model])</code>	Single OR double in one function: supplying <code>upperBarrier</code> selects a double. <code>model</code> = ANALYTIC (default, exact) or LSV (single-barrier only — a double under LSV is rejected).
<code>CELNET.WIN</code> <code>DOWBARRIER</code>	<code>(pair, tenor, strikeOrDelta, callPut, notional, barrier, [side], [windowStart], [windowEnd], [mcPairs], [mcSteps], [mcSeed])</code>	Partial-time knock-out active only in <code>[windowStart, windowEnd]</code> (no closed form). <code>mcPairs > 0</code> selects MC; <code>0</code> selects the exact ADI-PDE engine.
<code>CELNET.DIG</code> <code>ITAL</code>	<code>(pair, tenor, strike, callPut, notional, [style], [payout])</code>	Binary; <code>style</code> = CASH_OR_NOTHING (default)/ASSET_OR_NOTHING. Exact closed-form, no std-error.
<code>CELNET.TOU</code> <code>CH</code>	<code>(pair, tenor, kind, barrier, notional, [rebate], [upperBarrier], [monitoring])</code>	OT / NT / DNT / DOT. DNT/DOT require a strictly-greater <code>upperBarrier</code> . Exact closed-form.
<code>CELNET.VAR</code> <code>SWAP</code>	<code>(pair, tenor, notional, [strikeVol])</code>	Variance swap; fair-variance strike by log-contract static replication. 2×N <code>[fair_variance, K_var] / [fair_vol, √K_var]</code> + footer.

FUNCTION	SIGNATURE	ENGINE / STD-ERROR
CELNET.VOL SWAP	(pair, tenor, notional, [strikeVol])	Volatility swap; convexity-adjusted fair-vol strike <code>K_vol</code> (strictly $< \sqrt{K_var}$).
CELNET.ASI AN	(pair, tenor, strike, callPut, notional, [averaging], [observations], [method], [elapsedAvg], [elapsedWeight])	Fixed-strike arithmetic-average Asian; <code>method</code> = CURRAN (default)/TW; the (<code>elapsedAvg</code> , <code>elapsedWeight</code>) pair prices a seasoned average. Analytic, no std-error.
CELNET.FOR WARDSTART	(pair, tenor, callPut, moneyness, reset, notional)	Forward-start vanilla; strike fixed at the future reset to <code>moneyness × S(reset)</code> . Closed-form, no std-error.
CELNET.CLI QUET	(pair, tenor, callPut, moneyness, periods, notional, [localFloor], [localCap], [globalFloor], [globalCap], [mcPairs], [mcSeed])	Ratchet. Plain (no clamp) = exact Σ forward-start legs; supplying any floor/cap makes it MC-priced. Clamps presence-tracked.
CELNET.QUA NTO	(pair, tenor, callPut, strike, notional, conversionVol, correlation, [payoff])	Payoff on the foreign underlying, settled in the fixed currency, with the underlying $\leftrightarrow$ settlement-FX correlation drift adjustment. <code>payoff</code> = VANILLA (default)/DIGITAL. Closed-form.
CELNET.TAR F	(pair, tenor, callPut, strike, target, leverage, fixings, notional, [redemption], [fixingNotional], [mcPairs], [mcSeed])	Target-Redemption Forward; always MC (breaching-fixing gap risk). <code>redemption</code> = FULL_GAIN (default)/CAPPED_GAIN. Std-error row present.
CELNET.ACC UMULATOR	(pair, tenor, pivot, barrier, leverage, fixings, notional, [monitoring], [fixingNotional], [mcPairs], [mcSeed])	Accumulator (<code>barrier > pivot</code> up-and-out); always MC . Std-error row present.
CELNET.LOO KBACK	(pair, tenor, callPut, notional, [style], [monitoring], [strike], [observations], [mcPairs], [mcSeed])	<code>style</code> = FLOATING (default)/FIXED; <code>monitoring</code> = CONTINUOUS (default, exact closed form)/DISCRETE (MC over <code>observations</code>).
CELNET.AME RICAN	(pair, tenor, strike, callPut, notional, [style], [bermudanSteps], [lsmPaths], [lsmExerciseDates], [lsmSeed])	American / Bermudan. Default = exact PSOR free-boundary FD (no std-error); <code>lsmPaths > 0</code> selects the Longstaff-Schwartz regression MC engine.
CELNET.BAS KET	(pair, tenor, callPut, strike, notional, legs, correlations, [kind], [mcPaths], [mcReplications], [mcSteps], [mcSeed])	Correlated multi-asset (<code>kind</code> = BASKET (default)/BEST_OF/WORST_OF). <code>legs</code> = a range, one row per leg; <code>correlations</code> = the N×N matrix. The server rejects a non-PSD matrix (never regularised). Always MC ; std-error present.

Roadmap note — CELNET.BASKET Greeks. The basket premium and its MC standard error are computed and returned today; the full multi-asset basket Greek strip (a per-leg $N \times \{\text{spot}, \text{vol}\}$ Jacobian plus cross-gammas) is delivered through Celnet's implementation program as a distinct increment on the pathwise/AAD machinery already proven for the vanilla and exotic legs. Until that increment lands the server returns a zeroed Greek strip beside the real measured premium — the value a client reads is always the engine's value, never a fabricated sensitivity.

Surface marking

FUNCTION	SIGNATURE	BEHAVIOUR / PHASE
CELNET.SURF ACE	(pair, tenor, [model])	Pulls the marked smile for a (pair, tenor); surfaces the smile model the server stamps as authoritative provenance, with the requested model shown alongside so a mismatch is visible. Spill: delta-pillar header, vol row, arb-status + footer.
CELNET.MARK SURFACE	(pair, tenor, model, atmVol, rr25, bf25, [rr10], [bf10])	Calibrates and marks under a chosen model; returns the calibrated smile + the assigned surface version so a later RFQ / PRICE can pin it. Supply (rr10, bf10) for a 5-point smile. Commits on evaluation.
CELNET.MARK K	(pair, tenor, pillar, vol, [model], [comment])	Contributes a single manual mark. Two-phase + idempotent: stages (PENDING); commits only on explicit task-pane confirmation. Server-side arbitrage gates apply. 1×3 status spill.

Five smile families on the wire — and in Excel. The model argument accepts **VV** (market hedge / Vanna-Volga), **SABR**, **SVI**, **SSVI**, and **eSSVI** (also accepted as **ESSVI** / **EXTENDED**). The Excel parser accepts all five case-insensitively, so the extended (eSSVI) surface is exposed through **MARKSURFACE** , **MARK** , and **SURFACE** exactly like the other four.

The task pane is the control surface. Connection, RFQ entry, and the confirmed contribute-mark — including the smile-model selector (VV / SABR / SVI / SSVI / eSSVI) that drives **MARKSURFACE** and **MARK** . The two-phase write means a recalc never silently publishes a price.

Streaming, server-side risk & observability

FUNCTION	SIGNATURE	BEHAVIOUR
CELNET.SU BSCRIBE	(pair, tenor, strikeOrDelta, callPut, notional)	Pins a live two-way to a cell, multiplexed on the single session and coalesced with identical-argument cells (refcounted; torn down on cell delete/recalc). Flips to (stale) / (resync) on a heartbeat gap.

FUNCTION	SIGNATURE	BEHAVIOUR
CELNET.SEVERIES	(pair, observable, [tenor], [delta])	Pins a live observable: ATM, SPOT, RR, BF, FWD — the same multiplex stream that drives the GUI's trend modes.
CELNET.RISK	(dimension, numeraire, [rates], [scope])	Hierarchical risk aggregate over the org cube. dimension = FIRM/TRADER/BOOK/DESK/PAIR/LOCATION/ENTITY; everything converted into the chosen reporting numeraire (a missing required rate fails loudly, never a silent leg drop). Node grid.
CELNET.POSITIONS	([scope])	Lists the entitled open positions the cube aggregates, each with org placement + attribution; clean empty-state row. Position grid.
CELNET.LIMITS	(scope, numeraire, [rates])	The limit-tree utilization + RAG for a scope node — each limit's cap / exposure / ratio / RAG / enforcement / headroom, computed server-side. Limit grid.
CELNET.STATUS	()	Live server observability: connection state plus the latest heartbeat's drain-side price latency (p50 / p99 / p99.9), the conflation-drop count, and the surface-version / correlation echo. Passive — never sends a request, safe to leave live.

These risk readers consume the firm's **server-computed** aggregate — the API-first rule: a client never loops positions and sums. The roll-up, entitlement-pruning, and reporting-numeraire conversion all happen server-side, so an Excel cell is bit-identical to the GUI Book/Risk views. Because there is exactly one contract and one pricing path, the spreadsheet is never a fork — a value computed in a cell, shown in the GUI, returned to the Rust SDK, or carried over FIX is the same value.

7.5 The Trader GUI — the Richest Client of the One Contract

The React/WebGPU front-end as just another client — five workspaces (Ticket / Stream / Surface / Risk / Book) driving the live contract, with no privileged path the other clients lack.

The Celnet trader GUI is a single-window desk application built on React with a WebGPU rendering path. It is **live by default**: on launch it opens a streaming WebSocket session to the engine and stays connected, so every number on screen is the server's value — bit-identical to the same call from the SDK, the CLI, or Excel. There is no privileged front-end path; the GUI is simply the most expressive client of the one Celnet contract.

The whole desk lives behind **one window and five workspaces** — Ticket, Stream, Surface, Risk, Book — reachable from a persistent left rail or by keyboard (`⌘1` ... `⌘5`). Workspaces never unmount as you switch: an in-progress scenario, a half-marked surface, or a pinned comparison survives every flip between lanes. Everything a trader needs to navigate — what slice of the firm, which pair, and any action at all — sits in the title bar and a global command palette.

Stream (⌘2) — the resting state of the desk. A living blotter of streaming two-way markets multiplexed over one streaming session; click a side to trade. Only changed numbers flash — calm under fire. Each row carries a per-row stream-health badge driven by real sequence and resync state.

The shell — one window, five workspaces

Three fixtures frame every workspace: a **left rail** with the five workspace glyphs (Ticket , Stream , Surface , Risk , Book ) , each labelled with its `⌘`-number; a **title bar** carrying the mark-less Celnet wordmark, the scope breadcrumb, the pair menu, and the `⌘K` search/command affordance (identity → scope → instrument → action); and a slim bottom **status ribbon** with live stream health, sequence number, lines healthy, LPs in competition, gap count, the active transport seam, a measured render-frame budget, the desk clock, and the real build stamp. The brand is coral-on-indigo with the Anaheim display face — no traffic-light chrome, no function codes, no `<GO>` .

The **scope breadcrumb** renders the current slice of the firm as a clickable path — Firm · Desk · Book · ... — wired through every data path in the GUI, so what you see in Stream, Risk, and Book is exactly the slice the breadcrumb names: the same server-side, entitlement-aware pruning the position-fact cube enforces, surfaced as a navigation control.

The pair navigator — two ways to move between pairs. The watchlist strip pins one tile per watched pair (label, live spot at pip precision, a tick cue, an inline sparkline); the title-bar pair menu is a real anchored dropdown of watched pairs whose footer escalates to the command palette for searching the entire universe — the right tool when a desk runs hundreds of pairs.

The command palette (⌘K) — the keyboard-first spine. Fuzzy-matches across three families at once: pairs (jump to any currency pair), workspaces (any of the five lanes), and actions — including promoting a structure straight into the live blotter (e.g. "Stream EUR/USD 1M 25Δ RR"). No function codes, no command syntax to memorise. A modal shortcuts overlay (?) reads its bindings from one source of truth so the advertised chords can never drift from the real ones.

Ticket (⌘1) — the analytics surface that is also the executable

Ticket (⌘1) — one card that is both the analytics surface and the order. Build any product in the catalogue from one structure selector, see its live two-way and full Greek vector and conventions on the face, and hit it without changing screens. The same card reaches the entire on-wire instrument set — vanilla and multi-leg strategies, first-generation exotics, structured / path-dependent products, Monte-Carlo structured (TARF, accumulator), American/Bermudan early exercise, and the correlated basket family.

Walking the card: a **header** with pair, structure selector, and a notional input labelled with the correct currency leg; a **tenor strip** (overnight through one year, plus IMM and an arbitrary broken date) naming each tenor identically to the Stream lane from a single shared formatter; a **legs / product block** that for vanilla and strategies shows side, call/put, the delta handle, and the resolved strike at the active pair's *real* market — never a stale constant — with a **Solve: zero-cost** chip for multi-leg structures, and a product-specific typed input block for the exotics (barrier levels, schedules, leverage, exercise style, basket legs and ρ). A **booking-model chip** routes the choice into the instrument's pricing-model directive — Default or Local-Stoch-Vol — appearing only when there is a real choice (the window barrier is locked to LSV). The **two-way** returns a live BID / MID / OFFER with a visible depleting last-look ring; **Sell** / **Buy** hit the bid or lift the offer (⌘↵ accepts the offered side). MC-priced products carry an explicit price standard error alongside the premium; var/vol swaps quote a fair strike. Once quoted, the full FX desk **Greek strip** renders beneath the two-way. **Stream this** ≈ and **Add to risk** 田 promote the *exact same* instrument object into other lanes with no re-keying.

Stream (⌘2) — the RFS blotter & click-to-trade

The Stream blotter is the desk's resting state, multiplexed over one streaming session. Each row carries the pair, structure (with the quoting owner when the wire attributes it), tenor, bid / premium-mid / offer, a trend cell, spot delta, implied vol, and a per-row stream-health badge. When a row is healthy and its maker token is still valid, the bid and offer cells are live buttons that send an execute against the row's short-lived, line-bound tradable token. A row of trend mode chips re-plots every row's trend column from a real streamed observable — Premium, ATM vol, 25Δ RR, 25Δ BF, Spot, Forward — each served from the contract's market-series feed.

Click-to-trade last-look in the blotter. A lifted line returns a last-look response bound to the exact streamed price, resolving to a typed Executed/Reject outcome — never a silent failure. The token is keyed-MAC, last-look bounded, and single-use, so a stale, forged, or already-consumed click is declined with a typed reason.

The decline-and-retry path. When a token has expired or been consumed, the click is cleanly declined and the trader re-requests against a fresh, re-validated line — the idempotent, last-look design means a retransmit can never double-deal and a stale token can never fill at a moved market.

Surface (§3) — mark and recalibrate

Surface (§3) — three linked views of one marked surface. A WebGPU-ready 3D mesh, the selected tenor's smile over delta, and a per-tenor marking grid with the five broker handles (ATM, 25RR, 25BF, 10RR, 10BF). Editing a handle recalibrates the surface live through the same calibration the server runs. Publish is disabled unless the *whole* surface is arbitrage-free (every tenor's butterfly and the cross-tenor calendar check). Five model chips — Market hedge, Stochastic vol, Parametric, Parametric-surface, and eSSVI — drive the mark-surface request; the family the server actually calibrated under is read back from the typed provenance the server returns and shown as "marked as ...".

Risk (§4) — the scenario grid

Risk (§4) — a real-reprice shock grid. Every cell is a real reprice via the surface scenario service, never a Taylor approximation. Swappable axes over the five shock factors (Spot, Vol, Rate-dom, Rate-for, Time), metric tabs (P&L, spot delta, vega), and the server's book-shaped decomposition — bucketed vega per (tenor, delta) pillar, the cross-gamma stencil, and the theta-roll. Arriving here by drilling from the Book, a **< Book** affordance returns to the desk-wide cube — Book and Risk are two zooms of the same position-fact cube.

Book (§5) — desk-wide aggregated risk

Book (§5) — the desk-wide aggregated picture. Sums the real per-position risk across every open position across every pair into one desk-level view — each position repriced at its own pair's market and scaled by its signed notional. Summary cards (net P&L, vega, gamma, theta), a per-pair breakdown where any row drills straight into Risk for that pair's largest contributing position (labelled "largest of N"), and a book-summed aggregate vega ladder. Positions whose pair has no marked market are disclosed and excluded rather than priced against a fabricated market.

Light and dark themes, a high-contrast mode, the keyboard cheatsheet, the trend modes, the scope breadcrumb, and the pair navigator are all conveniences layered over the *same* API the Rust SDK, the CLI, and the Excel **CELNET.*** add-in consume. A value priced in the Ticket — across the full catalogue from vanilla through American/Bermudan and correlated baskets — streamed in the blotter, marked in the Surface under any of the five

smile families, shocked in Risk, or aggregated in the Book is the engine's value, identical across every client. The GUI is the desk's richest window onto Celnet, not a parallel implementation of it.

8. CelNet Estate Integration, Deployment Modes & Engineering Rigor

How Celnet slots into a running CelNet estate behind clean seam traits across three deployment modes — and (\$8.6 onward) the wall of executable proofs that earns every claim in \$1 through \$7.

Celnet is the **FX-options pricing system-of-record** inside the CelNet trade lifecycle. The broader CelNet estate routes, risks, clears and books FX flow; Celnet owns the option mathematics — price plus the full desk Greek set, the marked surface across all five smile families (VV · SABR · SVI · SSVI · eSSVI), the full on-wire product catalogue, and server-side hierarchical option risk — and injects them into both the price path and the order path through a small set of clean seam traits. The competitive consequence is decisive: a desk gets a front-to-back FX-options capability that slots into a *running* CelNet estate without standing up a parallel pricing stack and without a rewrite, behind one unversioned contract reachable identically from GUI, SDK, CLI, Excel and the WebSocket mirror.

The integration is built as **traits and adapters, exercised against real loopback servers and sinks**. The live binding to the running JVM estate is delivered through Celnet's implementation program: it slots in behind the already-built seams and is lit up against the running estate at deploy — an integration step on a finished interface, not new engineering.

Celnet in the CelNet trade lifecycle. Celnet supplies price, Greeks and the calibrated surface into the price path (`celnet-marketdata` → `marketmerchant` → distributor → CelNet Trader) and option risk into the risk and position path (`celnet-orderrouting` → `celnet-risk` → `celnet-destination` → `celnet-positionmanager`). The surrounding CelNet services own routing, credit, execution, clearing and the net-position truth source; Celnet owns the option intelligence at each point it is needed.

8.1 Where Celnet plugs in

The estate runs two complementary flows, and Celnet contributes to each at exactly the point where option intelligence is required.

Price path. Venue quotes arrive over FIX, are normalized by `celnet-marketdata`, assembled by `celnet-marketmerchant`, fanned out by the distributor over WebSocket, and rendered to the trader in **CelNet Trader** — the `celnet-orderrouting-webtrader-react` FX front-end. Celnet enriches this path with FX-option price, the full desk Greek set (price + 13 Greeks), and the live calibrated surface, so every option line the trader sees carries Celnet's marks, surface-version pinned.

Order path. From CelNet Trader an order flows through `celnet-orderrouting` (the order-lifecycle hub), `celnet-risk` (pre-trade credit and limit checks), `celnet-destination` (FIX-out to many venues), `celnet-clearing`, and into `celnet-positionmanager`, the net-position and P&L truth source. Celnet feeds option risk — sensitivities, FRTB-SA capital shape, and scenario surfaces — into the risk and position legs so credit checks and the position book reflect *true* option exposure rather than a notional approximation. Because aggregation is server-side and hierarchical (entitlement-pruned, numeraire-converted), the risk and position legs consume a firm roll-up rather than looping over Greek grids.

CELNET STAGE	ROLE	CELNET CONTRIBUTION
Market-data ingest & quote assembly	Normalize and merchant venue quotes	FX-option price, Greeks, calibrated surface into the quote
Distribution → CelNet Trader	Stream prices to the React front-end	Option marks on every line, surface-version pinned
Order-routing hub	Order lifecycle	Option valuation and risk on the working order
Pre-trade risk	Credit & limit gate	Option risk sensitivities + FRTB-SA capital for accurate exposure
Position manager	Net position & P&L truth	Option Greeks and marked-surface valuation, server-aggregated

8.2 The integration edge: FIX, governed egress, resilient ingest

Celnet meets the estate through a small, well-defined set of connectors, every one a real shipping component.

FIX 4.4 engine. A complete, hand-rolled, zero-copy FIX 4.4 implementation with the FIXT/4.4 session state machine (logon / heartbeat / test-request / resend / gap-fill), **both** session roles — an **acceptor** (quote publisher with quote-validity last-look) and an **initiator** (price-taker / hedge) — and an FX-options **dialect** (FXVO / FXNO; put/call / strike / exercise style / cut, convention-mapped). It carries quotes inbound and orders/executions outbound over the same protocol the rest of the estate and its venues already speak, and is gate-tested over a real loopback socket.

Conflating egress governor. Outbound price distribution passes through an egress governor that conflates to the latest value, paces with a token bucket, and **counts what it drops** — so a fast core never overwhelms a slower consumer (the distributor mailbox or a native socket), and conflation is observable rather than silent. Under the edge this governor sits over a lock-free **broadcast fan-out ring** that fans each pair's tick to all subscribers with zero-alloc publish and exact skip-accounting conflation.

Resilient market-data subscriber. A resilient WebSocket subscriber consumes upstream prices with reconnect, resubscribe, and sequence-gap resynchronization built in (subscribe → snapshot → sequenced deltas → on-gap full resync), so a transient feed interruption heals automatically instead of stalling the desk or delivering out-of-order updates.

Vendor-feed normalization, blend & divergence. A normalization layer maps heterogeneous upstream quotes into Celnet's canonical observables, blends multiple sources with staleness weighting, and surfaces divergence between them — gating out a stale or mispriced feed rather than silently averaging it. The blend / staleness / divergence **algorithm** is built and gated today; live multi-vendor quote **feeds** are delivered through Celnet's implementation program on the same normalization seam.

8.3 The order & execution lifecycle

The order path is fully realised at the integration edge. A streaming quote carries a single-use **tradable token**: an unguessable, keyed-MAC binding over the line (BUY@offer / SELL@bid) with a last-look validity window, minted on

the streaming session and redeemed by an idempotent execute. The same keyed-MAC path drives click-to-trade in the GUI and the live FIX acceptor, so the order-acceptance discipline — accept only a fresh, key-bound, not-yet-consumed token; reject anything stale, forged or replayed — is one mechanism behind one contract, not a per-client reimplementations.

Why one token path matters. A forged token, a replayed token, and a stale token are all rejected identically whether the order arrives from CelNet Trader's click-to-trade, the Rust SDK, or an inbound FIX order — because there is exactly one keyed-MAC mechanism, gate-tested against forged / replayed / expired inputs. The order path the estate sees is the order path the GUI sees.

Click-to-trade, keyed-MAC last-look. A streamed quote carries a single-use tradable token bound to its line and side, with a live last-look validity window. The same token discipline drives GUI click-to-trade, the SDK execute call, and the inbound FIX acceptor — accept fresh and key-bound, reject stale, forged or replayed.

Last-look expiry and retry. When the validity window lapses the execution is rejected cleanly and the client re-quotes for a fresh token — the resilient-resync discipline of §6.2 carried through to the order leg, so an expired token never books a stale price.

8.4 Three deployment modes, one codebase

Integration depth is a configuration choice, not a fork. Three seam traits — the **market-data source**, the **price sink**, and the order/execution adapter — are swapped to move Celnet along a spectrum from fully self-contained to fully estate-native. The engine core never names a transport: it is handed a market-data source and a price sink, and the edge builder picks the concrete adapters for the chosen deployment mode. Migration between modes is a reversible adapter swap, never a rewrite.

Standalone, Hybrid and CelnetIntegrated deployment modes. The same engine and the same contract sit behind three adapter configurations on the market-data / price-sink / order-and-execution seams. Standalone is fully real and gate-tested in-crate; CelnetIntegrated is the estate-native binding, with the live JVM/WS distributor wiring delivered through Celnet's implementation program on the same seam.

MODE	MARKET-DATA SOURCE	PRICE & ORDER SEAMS	USE
Standalone	Self-contained	Self-contained	Celnet runs as an independent FX-options pricing and risk platform — fully real and testable in isolation; the default for a self-contained tenant
Hybrid	Estate and external vendor feeds (blended)	Governed egress	Estate market data blended with an external vendor feed, Celnet-owned pricing and distribution
CelnetIntegrated	CelNet distributor feed	Governed publish to the CelNet distributor	Celnet as the estate's FX-options pricing system-of-record

Proven today, delivered through the program. The **Standalone** mode is fully real and gate-tested — its self-contained adapters record and replay every update. The **CelnetIntegrated** and **Hybrid** modes are wired through the *same* market-data-source / price-sink seam and exercised against a **real loopback server and sink** — proving the framing, the governed-egress conflation, and the resync contract. The estate-native binding itself — the JVM distributor sidecar handshake, mailbox calibration, and live quote-feed entitlement — is delivered through Celnet's implementation program: it slots in behind the already-built seam and is lit up against the running estate at deploy, an integration step on a finished interface. Because each mode is just a different binding of the same seam traits, a desk can start Standalone, prove the platform, and migrate deeper into the estate at its own pace.

8.5 Adaptable by design

The same market-data seam that ingests the CelNet distributor ingests external products and feeds. Celnet's normalization layer is a vendor-neutral **data-shape adapter** built to adapt to the venues and terminals a desk already runs — for example **Fenics**, **Bloomberg**, **Refinitiv** and **EBS**, which are integration/adaptor targets. Each is brought in as an adapter behind the seam that maps its quotes into the canonical observable set driving quote assembly, surface marking and live trend series. New sources are added as adapters behind the seam, so onboarding a feed is an integration, not a re-engineering of the pricing core.

The throughline across all of this is the platform's single principle: **one clean contract, consumed identically everywhere**. Whether Celnet runs Standalone or wired into the full CelNet lifecycle, the price the order path sees, the risk the position manager books, and the number on the trader's screen are the same value from the same source. Excel is a first-class client of that same contract — every **CELNET.*** function reaches the identical server, so a structuring desk and a risk desk price from one source of truth.

Excel as an estate client. The **CELNET.*** task pane consumes the one unversioned contract — smile-model selection across all five families (VV / SABR / SVI / SSVI / eSSVI) — so a spreadsheet desk prices from the same server and the same marks as CelNet Trader and the SDK.

Branded grid, contract-faithful. CELNET.* functions spill premium, Monte-Carlo standard error, the full Greek set and a convention footer — the same values, conventions and std-error discipline the estate and the GUI receive, never a spreadsheet re-implementation of the maths.

8 (cont.) · Engineering Rigor & Assurance

The wall of independent, executable proofs that runs on every change — the verification gate, per-class oracles, property/mutation/fuzz testing, the parity matrix, determinism, observability and supply-chain cleanliness — and the closing statement of the foundation of trust.

A pricing platform is only as trustworthy as the evidence behind its numbers. Celnet is engineered so that every headline claim — correctness, determinism, latency, resilience — is *earned and continuously re-proven*, not asserted. The differentiator is not a single clever gate; it is that the **whole platform ships behind a wall of independent, executable proofs that run on every change**, with each product class validated by the *strongest oracle that class admits*. Where an incumbent ships a number and a brochure, Celnet ships the number, the independent re-derivation that pins it, and the gate that fails the build if the two ever diverge.

8.6 The verification contract

Every change passes one gate before it is "done": a single command runs formatting, lint (warnings treated as errors), the full test suite, and license/advisory scanning. Numerical code is validated against references — never merely asserted plausible. The gate is the contract: a change that regresses a price, a property, a latency budget, a licence, or a parity claim fails the build before it reaches a desk.

8.7 Correctness, validated against the strongest oracle each class admits

Celnet does not make one blanket accuracy claim across the whole catalogue — different product classes admit different gold standards, and the platform applies the right one to each.

Analytic products — frozen golden tables. Vanilla prices and the full FX desk Greek set, both digital styles, all eight barrier variants, and one-touch values are validated to machine precision against **frozen QuantLib 1.42.1 reference tables** — independent maths, an independent implementation, agreeing to the last bits. The Heston (1993) stochastic-volatility European reference is gated identically against **hand-pinned published-literature values**, with provenance stated alongside the data. Beyond the tables, every Greek is **finite-difference cross-validated**: each of price, spot- and forward-delta, gamma, vega, theta, both rhos, vanna, volga, charm, speed, zomma and color must match its numerical bump-and-revalue twin.

Closed-form structured products — exact limits and published constants. The structured catalogue (variance/volatility swaps, arithmetic and geometric Asians, forward-start/cliquet, quanto, lookback) is pinned by the *exact analytic limit it must collapse to* plus a code-disjoint independent re-derivation — a variance swap's log-contract replication against an independently coded adaptive-Simpson quadrature and the `K_var ==  $\sigma^2$` flat-vol limit; a geometric Asian against Kemna-Vorst to $\sim 1e-12$. Regulatory capital is pinned the same way: the FRTB-SA low-correlation scenario transform is gated against **hand-computed BCBS constants** (`$\rho_{low} = \max(2\rho-1, 0.75\rho)$` , MAR21.6(2)), a defence added precisely because a longhand oracle that re-derives the same formula is a circular self-check — the BCBS-pinned constant breaks the circle.

Monte-Carlo and early-exercise products — explicit std-error bands. TARFs, accumulators, discrete-observation lookbacks, correlated baskets / best-of / worst-of (Cholesky MC), and American/Bermudan via Longstaff-Schwartz LSM carry a **reported price standard error** and are validated to converge *within their own stated stderr* of a code-disjoint independent MC. The first-generation exotics retain an internal triangulation gate on top: the

analytic reflection method, the Crank-Nicolson PDE with a Rannacher start-up, and the Philox Monte-Carlo path engine must agree — three independent numerical routes converging on one price.

The smile and surface engine carries its own correctness gates across all five families (VV / SABR / SVI / SSVI / eSSVI) — butterfly density non-negativity (a pointwise Breeden-Litzenberger re-pricing), calendar total-variance monotonicity, and vertical no-arbitrage — enforced as the surface is calibrated, so a marked surface that violates no-arbitrage is rejected at the source.

PRODUCT CLASS	STRONGEST ORACLE APPLIED	BAR
Vanilla, digital, barrier x8, touch	Frozen QuantLib 1.42.1 golden tables	Machine precision (last bits)
Heston European	Hand-pinned published-literature golden table	Machine precision (last bits)
Every Greek (analytic)	Finite-difference bump-and-revalue twin	Analytic $\approx$ numerical
Var/vol swap, Asian, fwd-start/cliquet, quanto, lookback	Exact closed-form limit + code-disjoint quadrature / Kemna-Vorst	$\sim 1e-6 \dots 1e-12$ to the limit
FRTB-SA capital	Hand-computed BCBS constants (breaks circular oracle)	Exact to the standard
TARF, accumulator, discrete lookback, basket, American (LSM)	Code-disjoint independent Monte-Carlo	Within reported price std-error
First-gen exotics (additional)	Analytic $\sim$ Crank-Nicolson-PDE $\sim$ Philox-MC triangulation	Three routes converge
Surface (VV / SABR / SVI / SSVI / eSSVI)	Butterfly / calendar / vertical no-arbitrage gates	No arbitrage admitted

8.8 Property, mutation and fuzz testing

Point checks confirm what an engineer thought to test; **property-based testing** confirms what they did not. Celnet asserts the invariants that must hold across whole input spaces — put-call parity, monotonicities, sign conventions, the branch behaviour of the strike $\leftrightarrow$ delta solver across the premium-adjusted call-delta maximum — and the harness searches for counterexamples rather than waiting for one in production.

Mutation testing then audits the tests themselves: the suite deliberately perturbs the implementation and confirms the tests catch the change. The gate spans the numerically load-bearing estate — the pricing core, the exotics catalogue, the surface engine, the risk cube, and counterparty-valuation adjustments — and **fails the build on any non-equivalent survivor**. The only exclusions are *provably semantically-equivalent* mutants, each justified inline and, for the solver cluster, verified against a 200k-case differential oracle that reproduces the unmutated result bit-for-bit.

Fuzz testing hardens the byte-level boundaries against adversarial input. Six fuzzing targets feed arbitrary bytes into the hand-rolled parsers and untrusted entry points — the pricing-input decoder, the journal recovery path, the

wire-contract decoder, and the three replicated-log decoders. Each target's contract is the same discipline: **never panic** (no slice overrun, no failure on an attacker-controlled length, no unbounded allocation), and return a value or a *typed* error — with re-encode soundness asserted on the success path.

8.9 An executable parity matrix

Capability claims are not prose in a brochure — they are codified as an **executable parity matrix** backed by roughly twenty-six integration checks, each gating a "meets or beats" claim against an **independent oracle**: eSSVI density and calendar arbitrage, variance/vol swaps, analytic Asians, Heston (Carr-Madan $\approx$ Fang-Oosterlee COS), forward-start/cliquet, LSV, Sobol-QMC variance reduction, baskets, FRTB-SA, the conventions/pair universe, GPU pathwise and path Greeks, the Raft election/snapshot/compaction proofs, and XVA. Every coverage, convention, surface, risk and streaming claim is exercised against the running system on every change, so the comparison against the vendor-neutral archetypes is *continuously proven rather than asserted*. If a capability regresses, the matrix fails the gate; the claim and the code never drift apart.

8.10 Deterministic, cross-platform behaviour

Determinism is a first-class guarantee. Celnet's counter-based random number generator is **bit-identical between CPU and GPU**, and GPU results are reconciled against a high-precision CPU oracle, so the same scenario produces the same number on the accelerator or the CPU SIMD fallback. Plugin execution is equally deterministic: the WebAssembly sandbox supports **bit-identical replay** across runs, the foundation for reproducible risk and auditable pricing. The distributed substrate inherits the same guarantee: a follower or recovered node replays the leader-replicated log to **bit-identical** state, with the workload deliberately containing a 1-ULP value so the bits must match.

(The GPU proofs establish *correctness and ratios* against the f64 CPU oracle on the M4 dev host, whose Metal backend lacks f64; the absolute NVIDIA throughput headline is delivered through the CUDA backend on NVIDIA hardware — the f64-correct kernels are already in place, so the program lights up the absolute figure on the target accelerator.)

8.11 Zero-cost observability

The platform is fully instrumented without taxing the hot path. Latency is captured in **tail-percentile HdrHistograms** that are coordinated-omission aware, and telemetry is offloaded over a **bounded, drop-on-full ring** so the pinned hot core stays log-, lock- and allocation-free. Operators get the p50/p99/p99.9 picture they need for mission-critical running — surfaced live on the stream heartbeat alongside conflation-drop counts, surface version and correlation id — while the maths pays nothing for it. Performance itself is held by **regression-gated benchmarks**: the in-core truth-gate asserts $p50 \leq 2\mu s$ / $p99 \leq 10\mu s$ / $p99.9 \leq 25\mu s$ **absolutely** (measured $\sim 42ns$ / $125ns$ / $\sim 1\mu s$ on the M4 dev host — a 24–80 $\times$ margin, host-local and single-core), and a machine-independent instruction-count gate runs in Linux CI.

8.12 Supply-chain cleanliness

Celnet is built exclusively on **open-source, permissively-licensed** software — no commercial libraries, solvers, or proprietary runtime data dependencies. The dependency set is policy-enforced in the gate: licences are checked

against an approved set, advisories are scanned, and banned or duplicate dependencies are rejected automatically. The result is a platform a bank can vet, deploy, and own end-to-end without commercial entanglement.

8.13 A resilient core: zero-allocation, hot-upgradable, durable, replicated

The runtime is engineered to keep trading through faults and upgrades. A **zero-allocation hot core** — wait-free single-producer/single-consumer rings feeding pinned cores, lock-free read-mostly state publication via an atomically swapped market-state snapshot and a single-writer seqlock top-of-book — does no allocation, locking or logging on the pricing path. **Blue-green zero-downtime state handoff** lets a new build take over from a running one without dropping the book, so hot upgrades are routine rather than an outage. Underneath, a **durable, checksummed, append-only journal** (with compaction/checkpoint built) provides clean crash recovery: a torn final record heals to the last good entry on restart, and the book and its repricing return bit-for-bit. Above the single node, a **leader-replicated event log with full Raft** (election + Pre-Vote, conflicting-tail truncation, snapshot compaction and InstallSnapshot reseed) commits only on quorum and replays to bit-identical state, proven on localhost multi-process.

How it all earns the claim. These mechanisms compose into a single discipline. Per-class oracle and finite-difference gates earn *correctness*; property, mutation and fuzz testing earn *robustness*; the parity matrix earns *competitiveness*; bit-identical RNG, plugin replay and Raft replay earn *determinism*; zero-cost observability and the absolute in-core truth-gate earn *performance*; the zero-allocation core, blue-green handoff, durable journal and replicated log earn *resilience*; supply-chain policy earns *ownership*. Built as a disciplined, layered Rust platform with an extensive automated suite, Celnet is a platform whose every number a desk can trade on, a quant can reproduce, and an architect can deploy — because the proof runs every time the code does.

8.14 The implementation program on the carry-generalized engine

A precise statement of the methods, architecture and seams proven today and the production absolutes Celnet's implementation program lights up on them. Every item below is fully seamed and adapter-ready on the engine; the program delivers each against the live target as an integration step on a finished interface, never a rewrite:

- **Cross-host wire p99 / SOTA kernel-bypass NIC datapath / absolute wire-latency SLOs** — delivered on production topology; the in-core truth-gate plus loopback benches establish the measured compute foundation.
- **CUDA/NVIDIA absolute GPU throughput, the $\leq 50\text{ms}$ exotic target, Workload-A/B absolutes** — delivered through the CUDA backend on NVIDIA hardware; correctness and ratios are proven against the f64 CPU oracle today (M4 / Lavapipe), with the f64-correct kernels already in place.
- **The live JVM CelNet estate lifecycle** (distributor sidecar handshake, mailbox calibration, live quote-feed entitlement) — lit up against the running estate at deploy; the seams and adapters (FIX 4.4 loopback, the deployment-mode swap) are built today.
- **Raft $\S 6$ dynamic membership / cross-DC transport / network-partition hardening** — delivered on production topology; correctness, quorum, election safety and framing are proven on localhost multi-process

today.

- **Plugin Tier-1 signed-shared-object and Tier-3 OS-sandbox** — seamed behind the frozen contract and delivered through the program; Tier-0 native and Tier-2 WebAssembly ship today.

This is the foundation of trust: Celnet labels every figure precisely by its measurement context (in-core / M4 / loopback) and presents every forward increment as a delivery step on a finished, state-of-the-art seam — methods, architecture and coverage that already out-cover the single-dealer incumbents, with the production absolutes lit up on the same engine. The platform is production-ready by construction: its seams, adapters and proofs are in place so that bringing it up against a running CelNet estate and accelerator fleet is integration, not new engineering.

